

Hierarchical DSTMs: Examples

The earlier chapters of this book have focused primarily on (hierarchical statistical) *models* for temporal, spatial, and spatio-temporal processes. In this chapter, we demonstrate, by way of example, the central role these models play in an “inference engine” that allows us to convert observations and information into knowledge. We do this for four examples based on spatio-temporal data and an underlying DSTM. Beyond these examples, the references given throughout this book contain many excellent applications of spatio-temporal statistical modeling. We encourage the reader to look at those sources to appreciate how much has been accomplished in quantifying and exploiting spatio-temporal variability in scientific applications.

Each of the four examples in this chapter was chosen to show different aspects of modeling, implementation, and inference. None of them represents a complete case study, although they are drawn from articles where the full details (from exploratory data analyses, to model building, to computation, to inference) are available. In Section 9.1, the example of long-lead forecasting of tropical Pacific sea surface temperatures is chosen to illustrate how a dimension-reduced BHM that includes a *linear DSTM* contrasts with the same for a particular *nonlinear DSTM*. The latter was built to account for possibly different transitions to and from “warm,” “normal,” and “cool” sea-surface-temperature regimes. Inferences based on the two BHMs’ posterior forecast distributions are used to compare their performances.

Section 9.2 presents a remote-sensing example, where very large datasets of aerosol measurements from the MISR instrument on NASA’s Terra satellite are analyzed. In this case, an EHM that includes a *linear DSTM* is fitted to the data, which compares directly to the case in Section 8.1 where a BHM with a linear DSTM is considered. Both use *dimension reduction*, but the computational aspects are quite different. The EHM requires parameters to be *estimated* separately, and in this section we give a method-of-moments estimator. Computing the EHM’s posterior means and variances is extremely

fast, relative to comparable BHM implementations, but inferences can be too liberal.

In Section 9.3, we consider an example of an invasive species, the Eurasian Collared Dove, that has been invading North America over the last several decades. This example was chosen to illustrate how a *non-Gaussian data model* can be paired with a *nonlinear DSTM* (process model) to obtain a scientifically motivated BHM of the species' invasion. Crucially, the parameter model in the BHM incorporates scientific information to account for uncertain probabilities of detection and landscape-scale variation in the spread and growth of the invasive-species population through time.

Finally, in Section 9.4, an example is considered of predicting and forecasting near-surface winds over the Mediterranean Sea based on remotely sensed scatterometer observations. The purpose of these predictions and forecasts is to provide realistic wind-forcing distributions for an operational ocean forecast model. We chose this example to illustrate how PDE-based scientific knowledge can be used to motivate a *multivariate linear DSTM*. In particular, we demonstrate how such a multivariate model can be built by conditioning on a common process. Another striking feature of this example is that it illustrates how the uncertainty captured in the BHM makes a *scientifically meaningful impact* on ocean forecasts.

This last example is a fitting conclusion to the last chapter, since we believe that the real potential of Statistics for spatio-temporal data can be found in modeling the *interaction of several (or many) variables* in space and time. The future of Statistics, in quantifying uncertainty realistically through multivariate hierarchical models and facilitating the ascent of the knowledge pyramid in Science, is bright!

9.1 LONG-LEAD FORECASTING OF TROPICAL PACIFIC SEA SURFACE TEMPERATURES

The process governing sea surface temperatures (SSTs) in the tropical Pacific Ocean is complex and influenced by interactions between the atmosphere and the ocean on a variety of temporal and spatial scales. In particular, its variability on interannual and interdecadal time scales is related strongly to changes in global-scale atmospheric processes (e.g., see the overview in Philander, 1990). On interannual time scales, the dominant features in this process are the so-called El Niño and La Niña phenomena, which are characterized by quasi-periodic warming and cooling that occurs with periods of approximately 3–5 years. In recent years, as observational networks and data-assimilation approaches have improved, the ability to forecast SSTs with lead times of 6–12 months has improved dramatically. Indeed, due to its complexity and uncertainties related to the fundamental physical mechanisms that drive these phenomena, SST is one of the few processes in oceanography in which “statistical” long-lead forecast models perform as well as or better than deterministic models (e.g., Barnston, Glantz, and He, 1999; van Oldenborgh et al., 2005).

State-space models in a reduced-dimensional space (e.g., based on EOF projections) have proven to be effective long-lead prediction models for quite some time (e.g., Penland and Magorian, 1993). Nevertheless, there is strong evidence that the dynamics of El Niño and La Niña are not linear (e.g., Hoerling, Kumar, and Zhong, 1997; Burgers and Stephenson, 1999; Kondrashov et al., 2005). In this regard, several nonlinear statistical models for SSTs have been developed in recent years, and their forecast performance has generally been comparable to or better than linear models, at least in terms of capturing the magnitude of the El Niño and La Niña events (e.g., Tangang et al., 1998; Berliner, Wikle, and Cressie, 2000; Tang et al., 2000; Timmermann, Voss, and Pasmanter, 2001; Kondrashov et al., 2005).

Given that SST observations have been used to illustrate many of the exploratory methods for spatio-temporal data in Chapter 5, it is appropriate to follow up in this chapter with long-lead forecasting models for SSTs based on hierarchical modeling. In what is to follow, we compare BHM implementations of linear and nonlinear DSTMs. In particular, our goal will be to forecast, six months in advance, tropical Pacific SST anomalies from current and past monthly averages. The data used in these examples are monthly averaged SST anomalies (where the anomalies are with respect to a January 1970–December 1985 climatology) on a $2^\circ \times 2^\circ$ resolution grid over the tropical Pacific Ocean from 29°S – 29°N latitude and 124°E – 70°W longitude. In reality, they are the result of combining *in situ* observations (e.g., buoys, ship tracks, etc.) with satellite observations and interpolating them onto the $2^\circ \times 2^\circ$ resolution grid; see (2.11) for a generic way to express this. We consider data for the period January 1970–January 1999 (349 consecutive months). There are 2261 oceanic locations in this dataset giving a total of 789,089 (2261×349) spatio-temporal data—a high-dimensional dataset. Figure 5.4 shows image plots for these data from February 1998 through January 1999. These plots illustrate the latter part of the extreme 1997 El Niño event, followed by the transition into the extreme 1998 La Niña event. For a more complete description of these data, see Chapter 5 and Berliner, Wikle, and Cressie (2000), *hereafter BWC*. In this section, we build a BHM based on a reduced-dimensional linear DSTM, and we compare it to one based on a reduced-dimensional nonlinear DSTM. We note that BWC chose a seven-month-ahead forecast to allow a month for the SST data to become available and for the forecast model to be run; in this section, we use six months lead time, in line with other forecasts.

9.1.1 Reduced-Dimension Linear DSTM

In this subsection, we consider a BHM for a reduced-dimension *linear* DSTM, to perform long-lead (six month) forecasting of monthly SST anomalies. Let the data be represented by $\mathbf{Z}_t \equiv (\mathbf{Z}_t(\mathbf{s}_1), \dots, \mathbf{Z}_t(\mathbf{s}_m))'$ for $m = 2261$ oceanic grid cells, and $t = 1, \dots, T = 349$ months. We let the underlying latent SST process be represented by $\mathbf{Y}_t \equiv (\mathbf{Y}_t(\mathbf{s}_1), \dots, \mathbf{Y}_t(\mathbf{s}_n))'$, where in this case $n = m = 2261$. Critically, we assume that the dynamics can be captured exclusively by the

reduced-dimensional process $\{\alpha_t\}$, where

$$\mathbf{Y}_t = \Phi \alpha_t + \gamma_t,$$

and Φ is an $n \times p_\alpha$ matrix of the first p_α EOFs of the empirical spatial covariance matrix of the data. The exploratory spatio-temporal data analysis in Chapter 5 was an important precursor to building this model; see the discussion in Section 5.3 and the associated plots of the first four EOFs in Figures 5.17 and 5.18. For simplicity, in this example we shall assume (somewhat unrealistically) that the error process has white-noise structure; that is, $\gamma_t \sim iid \text{Gau}(\mathbf{0}, \sigma_\gamma^2 \mathbf{I})$. We seek to forecast the smoothed process, $\mathbf{Y}_{t+\tau}^P \equiv \Phi \alpha_{t+\tau}$, given data through time t (i.e., $\mathbf{Z}_{1:t} \equiv \{\mathbf{Z}_1, \dots, \mathbf{Z}_t\}$), where $\tau = 6$ (corresponding to a forecast with 6-month forecast lead time). It is important to reiterate that in this application we are focused on *forecasting*; and since we are assuming that the dynamics are controlled by $\{\alpha_t\}$, it is reasonable to consider only the smoothed process $\{\mathbf{Y}_t^P\}$. We note that if the goal was spatio-temporal prediction, then we would want to take into account the truncation error, γ_t , and forecast $\mathbf{Y}_t \equiv \mathbf{Y}_t^P + \gamma_t$ (e.g., see Wikle and Cressie, 1999; Cressie, Shi, and Kang, 2010).

Consider the data model,

$$\mathbf{Z}_t = \mathbf{Y}_t^P + \mathbf{v}_t, \quad \mathbf{v}_t \sim iid \text{Gau}(\mathbf{0}, \Sigma_v), \quad (9.1)$$

where $\mathbf{Y}_t^P$ (i.e., α_t) and not $\mathbf{Y}_t$ is featured, and the error process $\{\mathbf{v}_t\}$ includes both the SST measurement/representativeness error (Section 7.1.2) and the truncation error (Section 7.1.3). In this implementation of a linear DSTM, we assume $\Sigma_v = \sigma_v^2 \mathbf{I}$.

The process model in this case is given by the p_α -dimensional process,

$$\alpha_{t+\tau} = \mathbf{M} \alpha_t + \eta_t, \quad \eta_t \sim iid \text{Gau}(\mathbf{0}, \Sigma_\eta), \quad t = 1, 2, \dots, \quad (9.2)$$

where we note that the α -process at time $t + \tau$ is conditioned on the α -process at time t to obtain the long-lead forecast [see BWC for relevant discussion].

Since we are taking a BHM approach, the parameter model must be specified; here, we choose

$$\begin{aligned} \text{vec}(\mathbf{M}) &\sim \text{Gau}(\mathbf{0}, 100 \mathbf{I}), \\ \Sigma_\eta^{-1} &\sim W((\mathbf{S}_\eta d_\eta)^{-1}, d_\eta), \quad \text{with } \mathbf{S}_\eta = 100 \mathbf{I}, d_\eta = p_\alpha - 1, \\ \sigma_v^2 &\sim IG(q_v, r_v), \quad \text{with } q_v = 0.1, r_v = 100, \end{aligned}$$

where $W(\cdot, d_\eta)$ corresponds to a Wishart distribution with d_η degrees of freedom, and the choices for q_v and r_v correspond to an inverse gamma distribution with mean, 0.1 and variance, 100. These prior distributions were chosen to suggest relatively non-informative priors, and a sensitivity analysis showed that the

results were not overly sensitive to these choices. Note that since we have SST data for quite a long time period ($T = 349$), we do not expect our results to be very sensitive to the initial conditions, so rather than specify prior distributions for the initial conditions, we simply fix $\alpha_t = \Phi'Z_t$; $t = 1, \dots, \tau = 6$.

For this BHM implementation, we use the basic Gibbs-sampler MCMC for linear DSTMs described in Section 8.3. However, rather than inferring a data-model error matrix that is left unspecified, recall we use $\sigma_v^2 \mathbf{I}$ here. Then, in the Gibbs sampler in Section 8.3, the full conditional distribution of σ_v^2 is given by

$$\sigma_v^2 \sim IG \left(\left(\frac{1}{r_v} + (1/2) \sum_{t=1}^T (\mathbf{Z}_t - \Phi \alpha_t)' (\mathbf{Z}_t - \Phi \alpha_t) \right)^{-1}, \frac{mT}{2} + q_v \right).$$

The algorithm was run for 6000 iterations, with the first 1000 iterations set aside as a period of “burn-in.” From trace plots, the algorithm appears to converge quite rapidly.

BHM Results: Linear DSTM

For this example, we considered two separate long-lead forecasts with $p_\alpha = 10$, corresponding to the leading EOFs in the reduced-dimension linear DSTM given by (9.2). First, we forecasted the six-month-lead October 1997 smoothed SST anomalies (i.e., $\mathbf{Y}_{t+\tau}^P$), given data through t , which corresponds to April 1997. Then we performed an equivalent forecast for October 1998, given data through April 1998. These forecasts are shown in Figure 9.1. In particular, the first row of panels, (a) and (f), show the data (i.e., $\mathbf{Z}_{t+\tau}$) for the forecast target months, respectively. The second row of panels, (b) and (g), show the smoothed data (i.e., $\Phi \Phi' \mathbf{Z}_{t+\tau}$) for the forecast target months; the third row of panels, (c) and (h), show the corresponding smooth forecasts (i.e., $E(\mathbf{Y}_{t+\tau}^P | \mathbf{Z}_{1:t})$); the fourth row of panels, (d) and (i), show the pixelwise 2.5 percentiles of the posterior forecast distribution (i.e., $[\mathbf{Y}_{t+\tau}^P | \mathbf{Z}_{1:t}]$), and the fifth row of panels (e) and (j) show the corresponding pixelwise 97.5 percentiles of the posterior forecast distribution (i.e., $[\mathbf{Y}_{t+\tau}^P | \mathbf{Z}_{1:t}]$).

Additionally, Figure 9.2 shows the posterior forecast density for SST averages over the Niño 3.4 region ($5^\circ \text{ S} - 5^\circ \text{ N}$, $120^\circ - 170^\circ \text{ W}$) for (a) the forecast of October 1997 (onset of an El Niño event) given data up to April 1997 and (b) the forecast of October 1998 (onset of a La Niña event) given data up to April 1998. The posterior means for these distributions are shown with a blue “star” on the horizontal axis, and the calculated Niño 3.4 index from the actual observations (6 months after the forecast was made) are shown with a red “star.” These forecast distributions are quite narrow, suggesting more accuracy than appears to be realistic, when compared to the observed values. In particular, although the forecast for October 1997 did indicate a higher (i.e., warmer) than average Niño 3.4 index, it did not come close to capturing the intensity of the actual warm (El Niño) event. In the case of the cool (La Niña) event in October 1998, the forecast distribution does contain

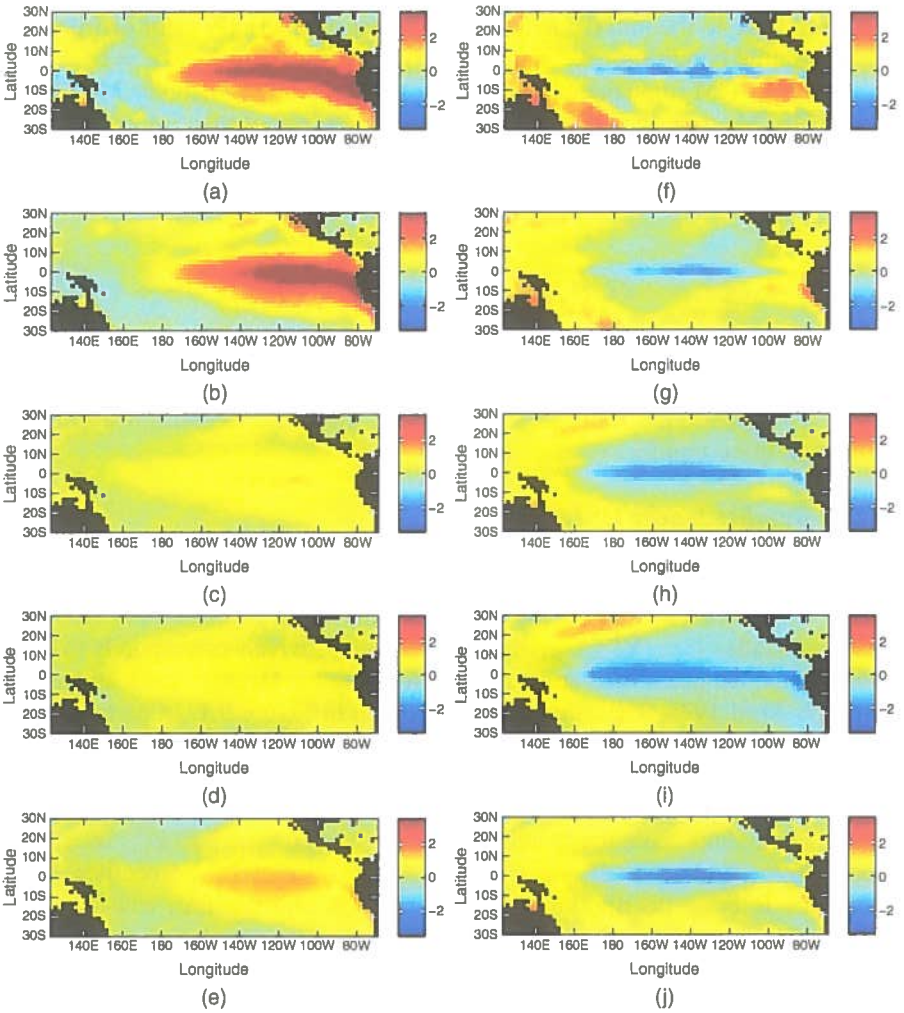


Figure 9.1 Sea Surface Temperature anomalies for October 1997 (left column) and October 1998 (right column). The first row of panels (a, f) show data for October 1997 and 1998, respectively. The second row of panels (b, g) show the smoothed data for October 1997 and 1998, respectively. The third row of panels (c, h) show the smooth forecast (mean of the posterior forecast distribution) from the reduced-dimension linear DSTM for October 1997 and October 1998, given data through April 1997 and April 1998, respectively. The fourth row of panels (d, i) show the pixelwise lower 2.5 percentiles from the posterior forecast distribution for October 1997 and 1998, respectively. The fifth row of panels (e, j) show the pixelwise upper 97.5 percentiles of the posterior forecast distribution for October 1997 and 1998, respectively.

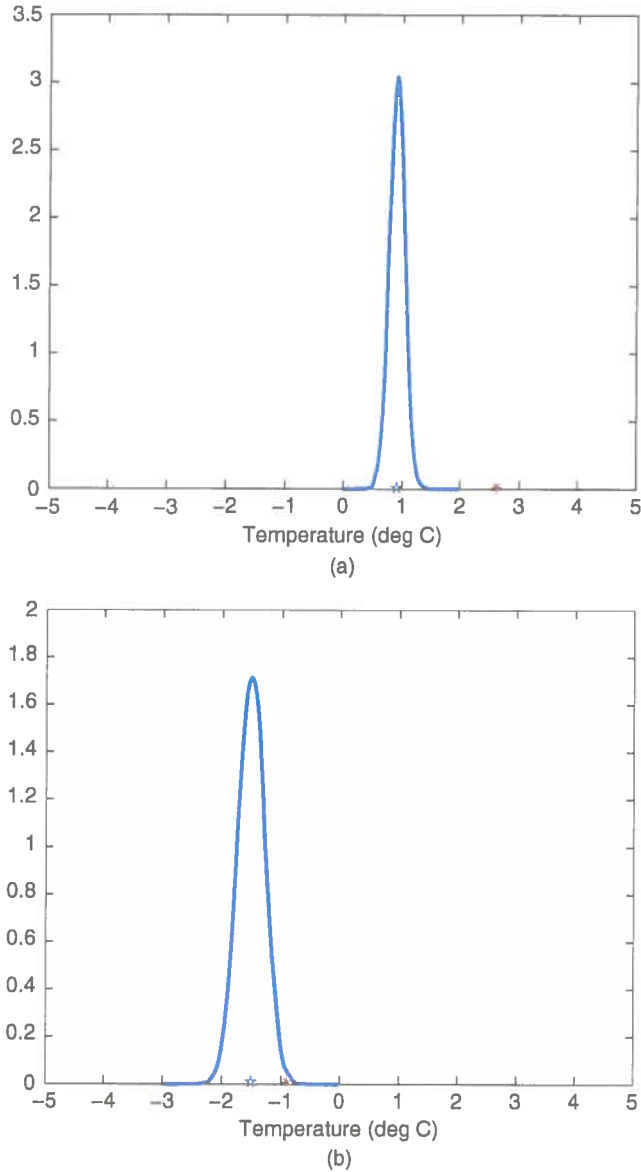


Figure 9.2 Reduced-dimension linear DSTM posterior forecast density of monthly Niño 3.4 region (5° S– 5° N, 120° – 170° W) SST averages. The top panel (a) shows the posterior forecast density for October 1997 given data up to April 1997. The bottom panel (b) shows the corresponding posterior forecast density for October 1998 given data up to April 1998. In each panel, the blue “star” shows the mean of the forecast distribution, and the red “star” shows the corresponding observed value (6 months later) of the Niño 3.4 region.

the actual observation, although the linear DSTM's forecast of the intensity of the cool event was somewhat extreme.

In summary, these SST BHM results show that the reduced-dimension linear DSTM did quite well at forecasting the onset of the strong 1998 La Niña. However, although it captured some warming associated with the preceding 1997 El Niño, it underestimated the magnitude of the warming quite dramatically. In addition, the forecast uncertainty was somewhat captured for the La Niña cooling, but not for the El Niño warming. One might question if the mixed success of the linear DSTM is the result of the (somewhat extreme) truncation to $p_\alpha = 10$ EOFs. Sensitivity analyses showed that keeping $p_\alpha = 20$ EOFs did not change substantially the posterior forecast distributions. This limited example for the 1997–1998 period suggests that at least the onset of the La Niña is reasonably represented by a linear DSTM; however, that is not the case for the onset of the El Niño.

9.1.2 Reduced-Dimension Nonlinear DSTM

BWC implemented a reduced-dimension nonlinear DSTM for long-lead forecasting of Pacific SST anomalies. The emphasis in that article was a model that allowed the dynamics of the El Niño and La Niña phenomena to be different (e.g., Hoerling, Kumar, and Zhong, 1997), due to a regime-switching model [e.g., see Section 7.3.2, equation (7.69)]. Given the complexity of this model, along with the fact that qualitative scientific knowledge was included in parameter specifications, it was natural to take a BHM approach. We reproduce the key elements of that analysis here.

The BHM in this context is similar to the reduced-dimension EOF model given in (9.1), with two major exceptions. First, rather than assuming $\Sigma_v = \sigma_v^2 \mathbf{I}$, we specify that $\mathbf{v}_t$ in the data model (9.1) is dependent in space; that is, $\mathbf{v}_t \sim iid \text{Gau}(\mathbf{0}, \Sigma_v)$, where BWC used a spatial covariance matrix Σ_v based on higher-order EOFs. More importantly, the process model on the reduced-dimensional space is

$$\alpha_{t+\tau} = \mu_t + \mathbf{M}_t \alpha_t + \eta_{t+\tau}, \quad (9.3)$$

where $\eta_t \sim iid \text{Gau}(\mathbf{0}, \Sigma_\eta)$, for all t . We also need to specify prior distributions for initial values $\{\alpha_t : t = 1, \dots, \tau\}$ in this framework (or, fix them as described in the linear DSTM given in Section 9.1.1.) The critical modeling assumption here is that we let $\mathbf{M}_t$ and μ_t be dependent on both the current (i.e., time t) and future (i.e., time $t + \tau$) climate regimes. That is,

$$\mathbf{M}_t = \mathbf{M}(I_t, J_t),$$

$$\mu_t = \mu(I_t, J_t),$$

where I_t classifies the current regime as “cool” (0), “normal” (1), or “warm” (2), and J_t anticipates a transition to one of the three regimes at time $t + \tau$.

Specifically, consider a threshold model for the index of the “current state”:

$$I_t = \begin{cases} 0, & \text{if } SOI_t < \text{low threshold,} \\ 1, & \text{if } SOI_t \text{ in between,} \\ 2, & \text{if } SOI_t > \text{upper threshold,} \end{cases}$$

where SOI_t is the Southern Oscillation Index (i.e., the surface air pressure anomalies at Tahiti minus those at Darwin), which is related to the El Niño and La Niña phenomena, and the threshold values are chosen empirically as described in BWC. The “future” state index J_t is then defined by the following latent process model:

$$J_t = \begin{cases} 0, & \text{if } W_t < \text{low threshold,} \\ 1, & \text{if } W_t \text{ in between,} \\ 2, & \text{if } W_t > \text{upper threshold,} \end{cases}$$

where W_t is a latent process that anticipates the future climate regime, with the threshold values calibrated empirically to the threshold parameters for I_t . This latent process is modeled by

$$W_t | \beta_w, \sigma_w^2 \sim \text{Gau}(\mathbf{X}_t' \beta_w, \sigma_w^2),$$

where

$$\mathbf{X}_t \equiv (1, U_t, U_t \sin(2\pi b_t/12), U_t \cos(2\pi b_t/12), U_t^2)',$$

U_t is the low-pass filtered east–west component of the wind at 10 meters above the surface at 5° N and 157° E, and b_t is an index of month (i.e., $b_t = 0, 1, \dots, 11$) at time t . Note that the use of the wind process at this location is motivated by known relationships between the atmospheric forcing and the ocean response in the Western Pacific ocean [see BWC for more discussion]. This relationship is evident empirically from Figure 5.8. However, there is recent evidence (SSES, 2009) to suggest that this relationship is changing.

The actual reduced-dimensional process model is somewhat more involved. In particular, we decompose the p_α -dimensional state process $\{\alpha_t\}$ into large-scale and small-scale components: $\alpha_t = (\alpha_t^{(1)'}, \alpha_t^{(2)'})'$, where $\alpha_t^{(1)}$ is a p_1 -dimensional vector corresponding to the first p_1 EOFs, and $\alpha_t^{(2)}$ is a p_2 -dimensional vector corresponding to the next p_2 EOFs (where $p_\alpha = p_1 + p_2$). Then, the evolution model,

$$\alpha_{t+\tau} = \mu(I_t, J_t) + \mathbf{M}(I_t, J_t)\alpha_t + \eta_{t+\tau},$$

is parameterized as

$$\begin{pmatrix} \alpha_{t+\tau}^{(1)} \\ \alpha_{t+\tau}^{(2)} \end{pmatrix} = \begin{pmatrix} \mu^{(1)}(I_t, J_t) \\ \mathbf{0} \end{pmatrix} + \begin{pmatrix} \mathbf{M}^{(1,1)}(I_t, J_t) & \mathbf{M}^{(1,2)} \\ \mathbf{M}^{(2,1)} & \mathbf{M}^{(2,2)} \end{pmatrix} \begin{pmatrix} \alpha_t^{(1)} \\ \alpha_t^{(2)} \end{pmatrix} + \begin{pmatrix} \eta_{t+\tau}^{(1)} \\ \eta_{t+\tau}^{(2)} \end{pmatrix}, \quad (9.4)$$

where

$$\begin{pmatrix} \eta_t^{(1)} \\ \eta_t^{(2)} \end{pmatrix} \sim iid \text{Gau} \left(\begin{pmatrix} \Sigma_\eta^{(1,1)} & \Sigma_\eta^{(1,2)} \\ (\Sigma_\eta^{(1,2)})' & \Sigma_\eta^{(2,2)} \end{pmatrix} \right). \quad (9.5)$$

As discussed in BWC, these parameterizations are motivated by the desire to make the model more parsimonious (i.e., to achieve dimension reduction in the parameter space).

At the next level of the HM hierarchy, we specify distributions for the parameters in the data model and the process model. In some cases, data-based hyperparameters were selected, with the variances inflated in the direction of non-informativeness. Considerable detail on specification of the parameter distributions can be found in BWC. More generally, the covariance matrix Σ_v defined in (9.1) was parameterized in terms of some of the truncated EOFs, as suggested in BWC and given by equation (7.26). The initial state-process components and propagator matrices were given multivariate Gaussian prior distributions with data-driven hyperparameters. Covariance matrices in the process model were given Wishart distributions with data-driven hyperparameters, and the regression parameters β_w and σ_w^2 in the model for W_t were given (data-driven) Gaussian and inverse-gamma priors.

SST BHM Results: Nonlinear DSTM

The MCMC algorithm for this model is given in specific detail in Berliner, Wikle, and Cressie (2000b). Although somewhat tedious to derive, it is quite efficient and consists exclusively of Gibbs updates.

One advantage of the threshold formulation described above for the process model, is that we can look at the posterior probability of the forecast state being in a particular regime j , with the long-lead forecast maps for that regime, and the probability-weighted mixture (the weighted mean over $j = 0, 1, 2$) of these regimes. That is, we can obtain

$$E(\alpha_{t+\tau} | \mathbf{Z}_{1:t}) = \sum_{j=0}^2 Pr\{J_t = j | \mathbf{Z}_{1:t}\} E(\mu(I_t, j) + \mathbf{M}(I_t, j) \alpha_t | \mathbf{Z}_{1:t}),$$

where recall that $\mathbf{Z}_{1:t}$ corresponds to all of the data up to time t . Hence, $E(\mathbf{Y}_{t+\tau}^P | \mathbf{Z}_{1:t}) = \Phi E(\alpha_{t+\tau} | \mathbf{Z}_{1:t})$. We can also obtain pixelwise percentiles of the posterior distribution, $[\mathbf{Y}_{t+\tau}^P | \mathbf{Z}_{1:t}]$, as well as summary measures of El

Niño/La Niña (e.g., Niño 3.4 values: averages over the 5° N to 5° S and 170° W to 120° W) and their posterior distributions.

We consider the same long-lead forecasting application described above in Section 9.1.1. As in that example, we consider $p_{\alpha} = 10$ EOFs in (9.4), with $p_1 = 4$, $p_2 = 6$. In contrast to that example, Σ_v in (9.1) used the next 20 EOFs to model spatial dependence. We ran the MCMC for 6000 iterations, the first 1000 of which we discarded as a period of “burn-in.”

Figure 9.3 shows the forecasts of the smoothed SST anomalies (i.e., $\mathbf{Y}_{t+\tau}^P$) for October 1997 and October 1998, given data through April 1997 and April 1998, respectively. As with the linear case shown in Figure 9.1, the first row of panels show the data (i.e., $\mathbf{Z}_{t+\tau}$) for the forecast target months, the second row of panels show the smoothed data (i.e., $\Phi\Phi'\mathbf{Z}_{t+\tau}$) for the forecast target months, the third row of panels show the corresponding smooth forecasts (i.e., $E[\mathbf{Y}_{t+\tau}^P|\mathbf{Z}_{1:t}]$), and the fourth and fifth rows of panels show the pixelwise 2.5 and 97.5 percentiles of the posterior forecast distribution (i.e., $[\mathbf{Y}_{t+\tau}^P|\mathbf{Z}_{1:t}]$).

Interestingly, when comparing these forecasts to those from the linear model, it is clear that the nonlinear model does a much better job of forecasting the intensity of the developing El Niño in October, 1997. However, the linear model seems to forecast the intensity of the developing La Niña better than the nonlinear model. Regarding variability, the spread of the nonlinear-model forecast distribution (as evidenced by the lower 2.5 and upper 97.5 percentiles) is quite a bit wider than that for the linear model, reflecting more realistic prediction uncertainty.

Additionally, Figure 9.4 shows the posterior forecast density for the Niño 3.4 region (5° S– 5° N, 120° – 170° W) SST averages for (a) the forecast of October 1997, given data up to April 1997, and (b) the forecast of October 1998, given data up to April 1998. The posterior means for these distributions are shown with a blue “star” on the horizontal axis, and the calculated Niño 3.4 averages from the actual observations (6 months after the forecast was made) are shown with a red “star.” As with the forecast fields, the nonlinear DSTM does a better job of forecasting the magnitude of the warm event than does the linear DSTM, but the nonlinear model does not forecast the cool event as well. However, the uncertainty bounds on the forecast are much wider for the nonlinear model, reflecting the fact that it accommodates more sources of uncertainty.

As mentioned above, the nonlinear DSTM allows forecasts to be considered that depend on the (unknown) regime of the hidden future state process. In addition, the posterior probability of each regime can be calculated. The predictions for each of these states (cool, normal, and warm) and their probabilities for the October 1997/October 1998 cases are included in Figure 9.5, along with their probability-weighted average. For the October 1997 (El Niño) forecast, the model assigned a 97% probability that the future state will be warm, thus weighting the prediction under that regime much more than the other two regimes (normal and cool). In contrast, the forecast for October 1998 (La Niña) assigned a 24% probability that the future state would be cool,

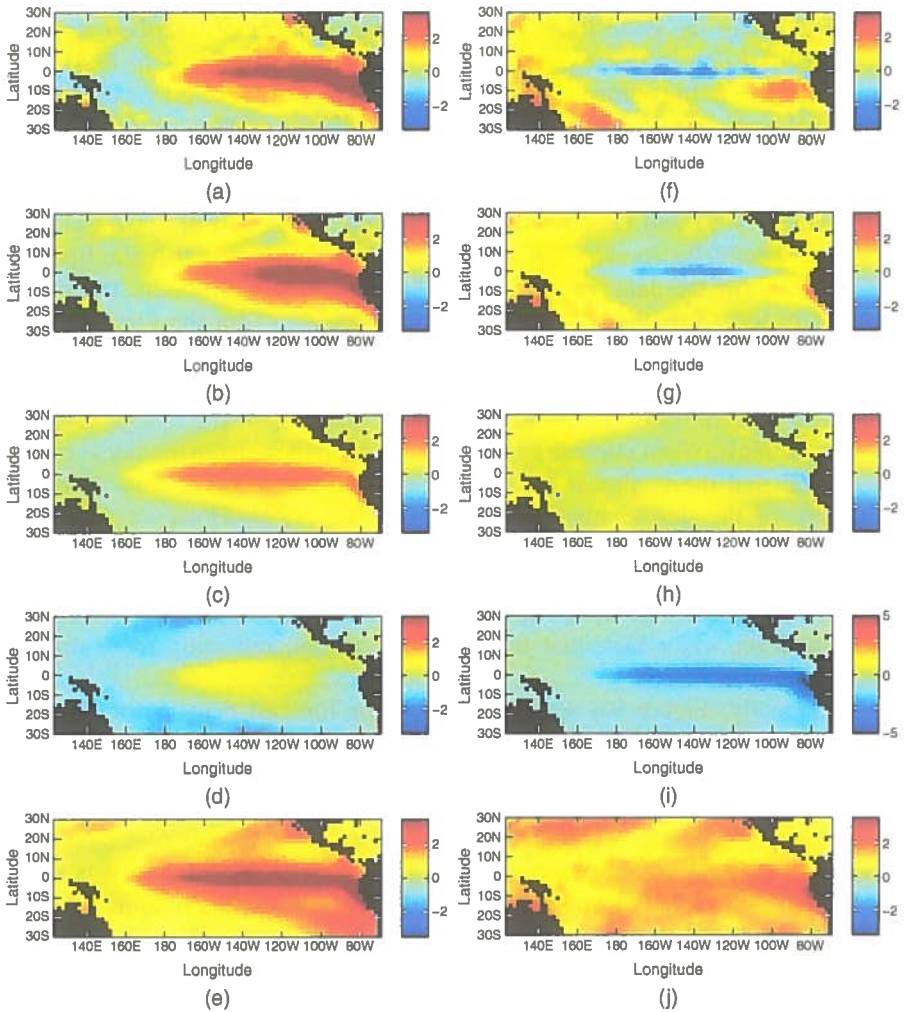


Figure 9.3 Sea Surface Temperature anomalies for October 1997 (left column) and October 1998 (right column). The first row of panels (a, f) show data for October 1997 and 1998, respectively. The second row of panels (b, g) show the smoothed data for October 1997 and 1998, respectively. The third row of panels (c, h) show the smooth forecast (mean of the posterior forecast distribution) from the reduced-dimension nonlinear DSTM for October 1997 and October 1998, given data through April 1997 and April 1998, respectively. The fourth row of panels (d, i) show the pixelwise lower 2.5 percentiles of the posterior forecast distribution for October 1997 and 1998, respectively. The fifth row of panels (e, j) show the pixelwise upper 97.5 percentiles of the posterior forecast distribution for October 1997 and 1998, respectively.

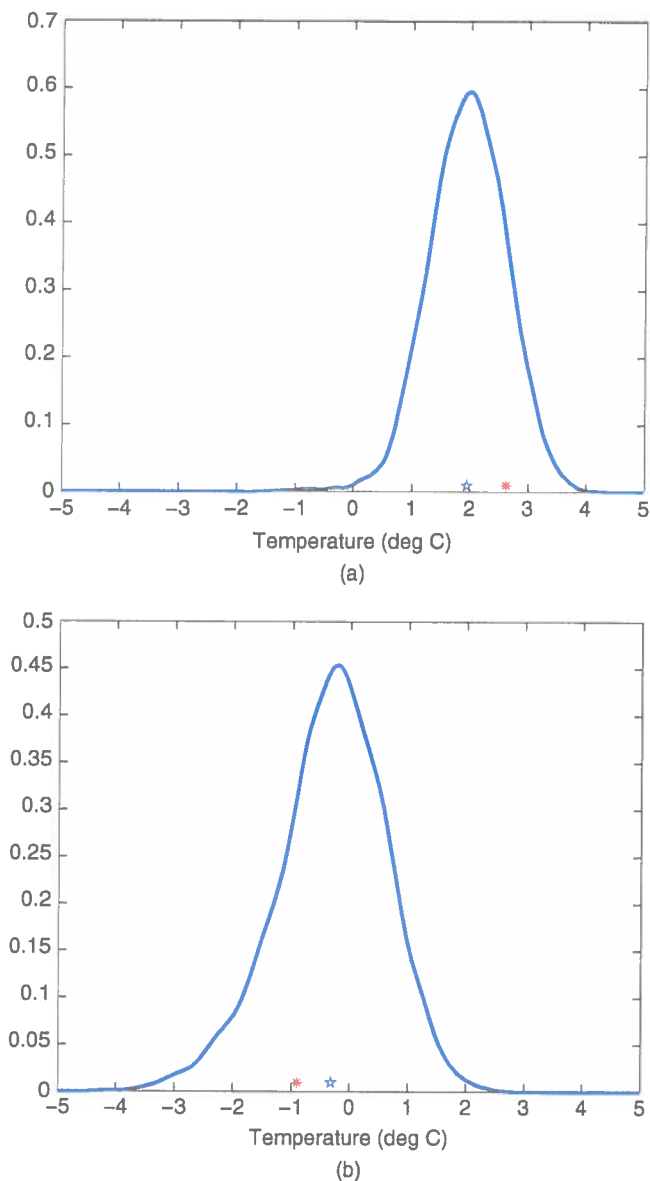


Figure 9.4 Reduced-dimension nonlinear DSTM posterior forecast density of monthly Niño 3.4 region (5°S – 5°N , 120° – 170°W) SST averages. The top panel (a) shows the posterior forecast distribution for October 1997, given data up to April 1997. The bottom panel (b) shows the corresponding forecast for October 1998, given data up to April 1998. In each panel, the blue “star” shows the mean of the forecast distribution and the red “star” shows the corresponding observed value (6 months later) of the Niño 3.4 region.

and a 74% probability that it would be normal. Thus, although the prediction associated with the cool state (Figure 9.5e) shows a substantial La Niña signature, the forecast assuming the normal state (Figure 9.5f), which has much higher posterior probability, shows that cooling is not as clear cut. Hence, the probability weighted average somewhat “washes out” the La Niña signature in this case. In principle, a forecaster could look at the forecasts for each regime and readjust these weights to produce a different forecast.

In conclusion, this example shows that there is a difference in the long-lead forecasts of the El Niño/La Niña phenomena during 1997–1998, as given by an SST BHM with a linear and a nonlinear DSTM. In addition to the differences in process dynamics, the linear and nonlinear implementations considered here differed in their assumptions about truncation-error variability: The linear BHM assumed truncation errors that were independent in time and space, while the nonlinear BHM assumed such errors were independent in time but spatially dependent (through additional EOF expansions). To be fair, we also considered the linear BHM with this same dependence structure in the truncation errors, and the forecasted results were not substantially different from those presented here, suggesting that the differences in model performance were due to the linear versus nonlinear dynamics in the DSTM. We note that many other types of nonlinear models could be considered for long-lead forecasting of SSTs. For example, Wikle and Hooten (2010) use the general quadratic nonlinear structure described in Section 7.3.3 to construct a BHM implementation of a nonlinear DSTM for long-lead forecasting of SSTs. Their results show that the 1997 El Niño can be forecasted quite well by such a model. (They did not implement their model to forecast the 1998 La Niña.) Wikle and Holan (2011) consider a quadratically nonlinear model for SST within the IDE framework. Their results corroborate the results presented here.

9.2 REMOTELY SENSED AEROSOL OPTICAL DEPTH

The National Aeronautics and Space Administration (NASA) has a remote-sensing program with a large array of satellites whose mission is earth-system science. To carry out this mission, NASA produces data at various levels: Level 2 data have been calibrated to the satellite’s footprint at high temporal resolution, although there is often a lot of missing data. Level 3 data are produced on a regular latitude-longitude grid over the whole globe at a coarser spatial and temporal (such as a day, a month, or a repeat-cycle of the satellite) resolution, and there are still substantial amounts of missing data.

NASA’s Terra satellite was launched on December 18, 1999, and the Multi-angle Imaging SpectroRadiometer (MISR) instrument on board collects global aerosol information; this includes aerosol optical depth, or AOD (a large value in a given pixel indicates a high aerosol content in the column of the atmosphere above that pixel), aerosol shapes, and aerosol sizes (Diner et al., 1999; Kaufman et al., 2000). MISR is one of the key instruments for long-term global aerosol monitoring. Since aerosol forcing is one of

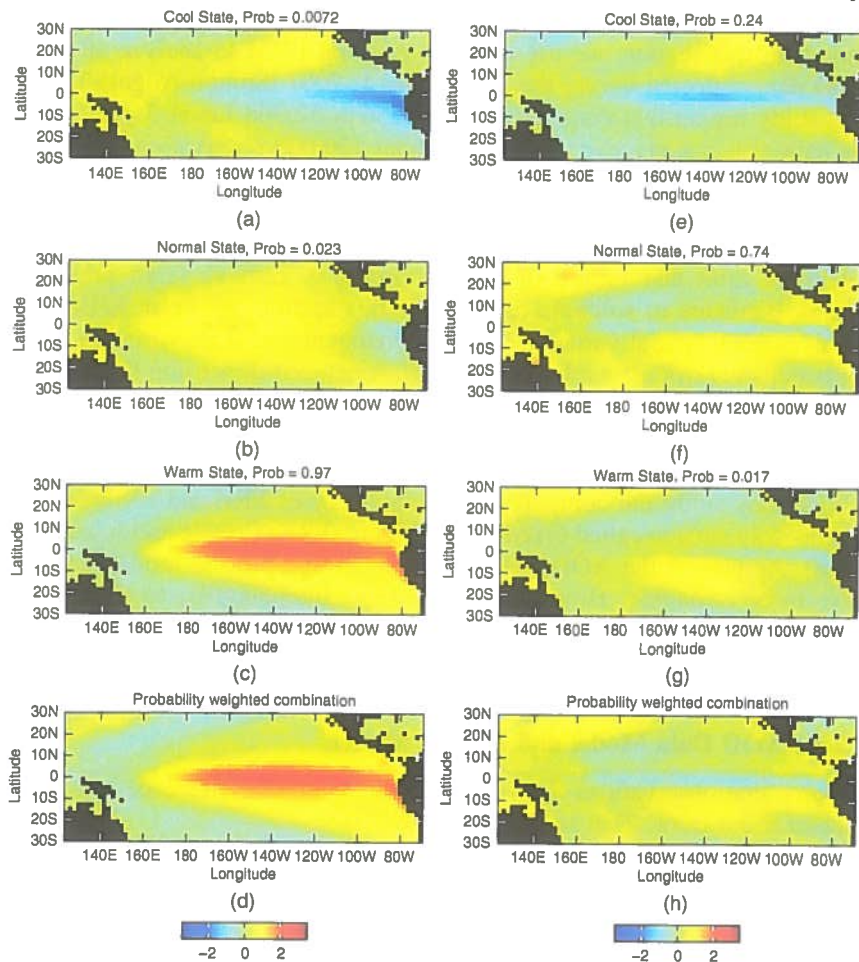


Figure 9.5 Nonlinear DSTM forecast of smooth SST anomalies for October 1997 (left column) and October 1998 (right column). Panels (a) and (e) show the forecast anomalies assuming that the future regime state in October 1997/1998 will be cold (regime 0); (b) and (f) show the forecast anomalies assuming the future regime state is normal (regime 1); (c) and (g) show the forecast anomalies assuming the future regime state is warm (regime 2). The title at the top of each panel gives the posterior probability, each to two significant figures, of these conditions. Panels (d) and (h) give the probability-weighted mixture of the images shown in the first three rows of panels for October 1997 and October 1998, respectively.

the major sources of uncertainty in climate models, accurate global aerosol records based on satellites play a vital role in calibrating the climate models. Consequently, complete-coverage aerosol values are needed.

Level 2 MISR data are collected at a spatial resolution of $17.6 \text{ km} \times 17.6 \text{ km}$. The satellite's orbit is repeated exactly, every 16 days. Level 3 data products are generated at a much coarser spatial resolution ($0.5^\circ \times 0.5^\circ$), by averaging Level 2 observations falling in the Level 3 pixels in a given time

period. In this section, we use a spatio-temporal EHM to analyze all Level 3 AOD data collected during the period July 1, 2001 through August 17, 2001, in a study region that consists of $128 \times 256 = 32,768$ Level 3 pixels. While the dataset is not massive, it involves about 120,000 observations and is large enough to demonstrate the power of dimension reduction in the AOD EHM given below.

The MISR AOD data are noisy and incomplete due to alignment of the satellite's orbit and failure of the instrument to retrieve (e.g., presence of clouds, hardware or software failures). In this section, we fit an EHM, where the dynamical spatial variation, the nondynamical fine-scale variation, and the measurement error are explicitly modeled. A reduced-dimension DSTM is used in the process model, and a rapid recursive updating procedure (i.e., a type of Kalman filter) is implemented to provide complete coverage of aerosol values. Cressie, Shi, and Kang (2010) call the methodology Fixed Rank Filtering (FRF); they study the computational complexity of FRF and compare it to a spatial-only version called Fixed Rank Kriging (FRK; Cressie and Johannesson, 2008). As part of the AOD EHM, data-model and process-model parameters have to be *estimated*. How this is done (e.g., likelihood based, loss-function based, moment based) is often determined by the practicalities of the problem; here, a method-of-moments approach is taken (Section 9.2.2).

9.2.1 AOD Data Model and Process Model

Assume that observations $Z_t(\cdot)$ are taken on $Y_t(\cdot)$, the difference being due to measurement error. That is, the *data model* is

$$Z_t(\mathbf{s}) = Y_t(\mathbf{s}) + \varepsilon_t(\mathbf{s}), \quad \mathbf{s} \in D_s, t = 1, 2, \dots, \quad (9.6)$$

where $Y_t(\cdot)$ is the underlying (true) spatio-temporal process that will be modeled at the process-model level of the HM; $\varepsilon_t(\cdot)$ is a mean-zero Gaussian process that is uncorrelated in space and time, and $\text{var}(\varepsilon_t(\mathbf{s})) = \sigma_\varepsilon^2 v_t(\mathbf{s}) > 0$. The only parameter in the data model is σ_ε^2 , since $v_t(\mathbf{s})$ is assumed known for all $\mathbf{s} \in D_s$ (e.g., see Section 9.2.3).

The *process model* for $\{Y_t(\mathbf{s}) : \mathbf{s} \in D_s, t = 1, 2, \dots\}$ assumes that the true (hidden) spatio-temporal process $Y_t(\cdot)$ can be decomposed as follows:

$$Y_t(\mathbf{s}) = \mathbf{X}_t(\mathbf{s})' \boldsymbol{\beta} + \kappa_t(\mathbf{s}) + \xi_t(\mathbf{s}), \quad \mathbf{s} \in D_s, t = 1, 2, \dots, \quad (9.7)$$

where the first, second, and third components on the right-hand side of (9.7) represent the large-scale, small-scale, and fine-scale components, respectively, of spatial variability. Define the known covariate vector, $\mathbf{X}_t(\cdot) \equiv (X_{1,t}(\cdot), \dots, X_{p,t}(\cdot))'$, and think of the p elements of $\boldsymbol{\beta}$ as parameters of the process model. That is, the large-scale component is assumed to be deterministic. In contrast, assume that $\kappa_t(\cdot)$ and $\xi_t(\cdot)$ are random processes that each have mean zero. Consequently, $E(Y_t(\mathbf{s})) \equiv \mu_t(\mathbf{s}) = \mathbf{X}_t(\mathbf{s})' \boldsymbol{\beta}$; $\mathbf{s} \in D_s$.

In the rest of this section, we assume that β and hence $\mu_t(\cdot)$ is known; in practice, this amounts to having a stable estimate of trend and then working with the detrended data. Although we lose a component of uncertainty due to estimation of the mean, the ultimate goal of a fast and efficient computational algorithm for very large data sets leads to this data/model/computing compromise (Section 2.5).

At any *fixed* time t , assume that $\kappa_t(\cdot)$ has mean zero and is written in terms of a basis function expansion (e.g., Cressie and Johannesson, 2008):

$$\kappa_t(\mathbf{s}) = \boldsymbol{\phi}(\mathbf{s})' \boldsymbol{\alpha}_t, \quad \mathbf{s} \in D_s, \quad (9.8)$$

where $\boldsymbol{\phi}(\cdot) \equiv (\phi_1(\cdot), \dots, \phi_{p_\alpha}(\cdot))'$ is a vector of p_α known *spatial* basis functions and $\boldsymbol{\alpha}_t \equiv (\alpha_t(1), \dots, \alpha_t(p_\alpha))'$ is a mean-zero Gaussian random vector with $p_\alpha \times p_\alpha$ covariance matrix given by

$$\text{var}(\boldsymbol{\alpha}_t) \equiv \boldsymbol{\Sigma}_t,$$

where we have chosen *not* to write this as $\boldsymbol{\Sigma}_{\alpha,t}$, for ease of exposition. Notice that $\text{cov}(\kappa_t(\mathbf{s}), \kappa_t(\mathbf{x})) = \boldsymbol{\phi}(\mathbf{s})' \boldsymbol{\Sigma}_t \boldsymbol{\phi}(\mathbf{x})$, which is a positive-definite function (Section 6.1.1) for any set of spatial basis functions chosen.

As time t progresses, we assume that $\{\boldsymbol{\alpha}_t : t = 1, 2, \dots\}$ evolves as a first-order linear DSTM process (Section 6.4):

$$\boldsymbol{\alpha}_{t+1} = \mathbf{M}_{t+1} \boldsymbol{\alpha}_t + \boldsymbol{\xi}_{t+1}, \quad t = 1, 2, \dots, \quad (9.9)$$

where $\mathbf{M}_{t+1}$ is a $p_\alpha \times p_\alpha$ propagator matrix that generally depends on time; and the p_α -dimensional Gaussian error vector, $\boldsymbol{\xi}_{t+1}$, is independent of $\boldsymbol{\alpha}_t$, has mean zero, and has covariance matrix, $\text{var}(\boldsymbol{\xi}_{t+1}) \equiv \mathbf{Q}_{t+1}$. Notice that in (9.9), we have chosen *not* to write the propagator matrix as $\mathbf{M}_{\alpha,t+1}$, nor the error covariance matrix as $\mathbf{Q}_{\xi,t+1}$, for ease of exposition. (The reason for this simplification in notation is that the model allows $\boldsymbol{\Sigma}_t$, $\mathbf{M}_t$, and $\mathbf{Q}_t$ to depend on t , and in the presentation that follows, double subscripts would be cumbersome.) Define the cross-covariances,

$$\boldsymbol{\Sigma}_{t,r} \equiv \text{cov}(\boldsymbol{\alpha}_t, \boldsymbol{\alpha}_r), \quad t, r = 1, 2, \dots, \quad (9.10)$$

where we have already notated $\boldsymbol{\Sigma}_{t,t}$ as simply, $\boldsymbol{\Sigma}_t$. Recall from Section 6.4.1 that for $r > t$ we have

$$\boldsymbol{\Sigma}_{r,t} = \mathbf{M}_r \mathbf{M}_{r-1} \cdots \mathbf{M}_{t+1} \boldsymbol{\Sigma}_t, \quad (9.11)$$

and

$$\boldsymbol{\Sigma}_{t+1} = \mathbf{M}_{t+1} \boldsymbol{\Sigma}_t \mathbf{M}_{t+1}' + \mathbf{Q}_{t+1}. \quad (9.12)$$

If we put $r = t + 1$ in (9.11), the $p_\alpha \times p_\alpha$ lag-1 cross-covariance matrix is given by

$$\Sigma_t^{(1)} \equiv \Sigma_{t+1,t} = \mathbf{M}_{t+1} \Sigma_t, \quad t = 1, 2, \dots \quad (9.13)$$

The cross-covariance matrix, $\Sigma_t^{(1)}$, captures the lag-1 (in time) cross-covariances between α_t and α_{t+1} , in the reduced-dimensional space.

The third component in (9.7), $\xi_t(\cdot)$, is assumed to be independent of $\{\alpha_t\}$, to have mean zero, and *not* to have temporally dynamical structure (as $\{\alpha_t\}$ does). Its role is to capture the fine-scale spatial structure, which in this application to AOD, is at the pixel scale. Here, we assume the simplest possible form for its covariance function, namely

$$E(\xi_t(\mathbf{s})\xi_r(\mathbf{x})) = \begin{cases} \sigma_\xi^2, & \text{if } \mathbf{s} = \mathbf{x} \text{ and } t = r, \\ 0, & \text{otherwise.} \end{cases} \quad (9.14)$$

All together, the parameters of the *process* model are: σ_ξ^2 , Σ_1 , and $\{\mathbf{M}_{t+1}, \mathbf{Q}_{t+1} : t = 1, 2, \dots\}$.

Upon combining the parameters of the data model and the process model, we obtain

$$\Theta \equiv \{\sigma_\varepsilon^2, \sigma_\xi^2, \Sigma_1, \{\mathbf{M}_{t+1}, \mathbf{Q}_{t+1} : t = 1, 2, \dots\}\},$$

which we assume are fixed but unknown. Their estimation is discussed in Section 9.2.2.

From (9.10), the covariance structure of the *spatio-temporal* process $Y_t(\cdot)$ is given by

$$\text{cov}(Y_t(\mathbf{s}), Y_r(\mathbf{x})) = \phi(\mathbf{s})' \Sigma_{t,r} \phi(\mathbf{x}) + \sigma_\xi^2 I(\mathbf{s} = \mathbf{x} \text{ and } t = r), \quad (9.15)$$

where $I(\cdot)$ denotes the indicator function. Furthermore, it is straightforward to see that the covariances between the process and the data are given by $\text{cov}(Y_t(\mathbf{s}), Z_r(\mathbf{x})) = \text{cov}(Y_t(\mathbf{s}), Y_r(\mathbf{x}))$, which is given by (9.15).

9.2.2 The AOD EHM: Implementation and Inference

Our principal interest is in inference on the process $\{Y_t(\mathbf{s}) : \mathbf{s} \in D_s\}$, given the data $\mathbf{Z} = (\mathbf{Z}'_1, \dots, \mathbf{Z}'_t)'$, which includes all of the observations up through time t (i.e., there are m_t observations for each data vector $\mathbf{Z}_t$). This is in contrast to Section 9.1, where our principal interest was in inference on $\{\Phi(\mathbf{s})' \alpha_t : \mathbf{s} \in D_s\}$, given the data $\mathbf{Z}$. Another difference is that an EHM approach is taken in this section, which involves *estimating* the parameters Θ and substituting them into the posterior distribution of $Y_t(\cdot)$.

Optimal Prediction

To make inference on $Y_t(\mathbf{s}_0)$, where $\mathbf{s}_0$ is a given location in D_s , we use the (estimated) posterior mean (Section 2.2.2), written here as $E(Y_t(\mathbf{s}_0)|\mathbf{Z}, \hat{\Theta})$. At first glance, this would require calculating $(\text{var}(\mathbf{Z}|\Theta))^{-1}$ and $\text{cov}(Y_t(\mathbf{s}_0), \mathbf{Z}|\Theta)$ in terms of Θ , and substituting $\hat{\Theta}$ for Θ in the formulas. These matrices are huge, due to both a large number of data locations, and because the latest time t is always increasing. The temporal aspect can be handled through jointly Kalman filtering $\{(\alpha_t, \xi_t(\mathbf{s}_0)): t = 1, 2, \dots\}$; the basic forecasting and filtering algorithm for $\{\alpha_t: t = 1, 2, \dots\}$ is given in Section 8.2.1, and Cressie, Shi, and Kang (2010) extend this to jointly filtering $\{(\alpha_t, \xi_t(\mathbf{s}_0)): t = 1, 2, \dots\}$. The spatial aspect is handled through dimension reduction and the Kalman filter's implicit incorporation of the Sherman–Morrison–Woodbury identity, (8.18); then all matrix inverses used to calculate the optimal predictor are either $p_\alpha \times p_\alpha$ or (large) diagonal matrices. Cressie, Shi, and Kang (2010) show that the computational complexity of calculating $E(Y_t(\mathbf{s}_0)|\mathbf{Z}, \Theta)$ and its mean squared prediction error, $\text{var}(Y_t(\mathbf{s}_0)|\mathbf{Z}, \Theta)$, is $O\left(\sum_{i=1}^t m_i\right)t$.

In terms of the notation in (9.6) and (9.7), if $\hat{\alpha}_{t|t}$ and $\hat{\xi}_{t|t}(\mathbf{s}_0)$ are the Kalman-filtered random effects, then the optimal predictor is

$$\hat{Y}_t(\mathbf{s}_0) = \mathbf{X}_t(\mathbf{s}_0)' \boldsymbol{\beta} + \boldsymbol{\Phi}_t(\mathbf{s}_0)' \hat{\alpha}_{t|t} + \hat{\xi}_{t|t}(\mathbf{s}_0), \quad (9.16)$$

which is called the Fixed Rank Filtering (FRF) predictor by Cressie, Shi, and Kang (2010). An expression for the mean squared prediction error matrix of $(\hat{\alpha}_{t|t}, \hat{\xi}_{t|t}(\mathbf{s}_0))'$, and hence the mean squared prediction error of (9.16), is given by Cressie, Shi, and Kang (2010). They call the square root of this latter quantity, which we write as $\sigma_t(\mathbf{s}_0)$, the FRF standard error. Then, estimates $\hat{\Theta}$ are substituted into $\hat{Y}_t(\mathbf{s}_0)$ and $\sigma_t(\mathbf{s}_0)$; as $\mathbf{s}_0$ varies over D_s , maps of predictions and associated prediction standard errors are generated. Since the variability of $\hat{\Theta}$ has not been accounted for, an EHM approach tends to result in inferences that are liberal. Again, this is a data/model/computing compromise (Section 2.5) that is critical for the model to process very large datasets in a relatively short time (on the order of minutes).

Parameter Estimation

Choosing to estimate parameters rather than putting a prior on them leads to choices about the estimation method to employ. There are two common types of estimators, one based on maximum likelihood and the other based on the method of moments. It is this choice and its implementation that is often the most labor-intensive part of implementation and inference in an EHM. In the AOD EHM, Θ is made up of scalar variance components, σ_ϵ^2 and σ_ξ^2 , and $p_\alpha \times p_\alpha$ matrices that are related through the expressions (9.11), (9.12), and (9.13); recall that $\boldsymbol{\beta}$ is assumed known. In what follows, we shall present estimation via the method of moments (e.g., Wikle and Cressie, 1999), although it should be noted that the likelihood-based EM algorithm could have been

tried (e.g., Xu and Wikle, 2007; Katzfuss and Cressie, 2011). The method-of-moments estimates we present have an advantage in that they are very fast to compute when datasets are very large.

We estimate σ_ε^2 and σ_ξ^2 separately, using empirical semivariograms for small spatial lags; the robust semivariogram estimator, $\hat{\gamma}(h)$, of Cressie and Hawkins (1980) was plotted against h , for h near the origin, and a straight line, $\tilde{\gamma}(h)$, was fitted by weighted least squares (Cressie, 1985). For regional data, Kang, Liu, and Cressie (2009) show that σ_ε^2 is estimated unbiasedly from the fitted line's intercept. Furthermore, σ_ξ^2 is estimated from another moment relation, this one involving the semivariogram at spatial lag 1 (Kang, Cressie, and Shi, 2010, Section 3).

Method-of-moments estimates of the $p_\alpha \times p_\alpha$ matrices, Σ_1 , $\{\mathbf{M}_{t+1}\}$, and $\{\mathbf{Q}_{t+1}\}$, are obtained by computing empirical covariance and cross-covariance matrices from the data $\mathbf{Z}_t$ and $\mathbf{Z}_{t+1}$ based on dividing D_s up into “bins” (spatial areas of larger support than the data). Each bin receives all the data in $\mathbf{Z}_t$ whose locations fall within the bin; likewise for $\mathbf{Z}_{t+1}$. The purpose of the bins is to average over them in order to decrease the variability of the estimated covariance matrix and lag-one cross-covariance matrix; hence, the number of bins will be considerably less than $\min\{m_t, m_{t+1}\}$. This suggests that the spatial basis matrix (Φ) must also be aggregated to account for the binning of the data. The details are given in Kang, Cressie, and Shi (2010), who show that the moment estimates can be obtained by minimizing appropriate Frobenius norms.

The estimation is done sequentially: Solve for $\hat{\Sigma}_1$ first, based on $\mathbf{Z}_1$ and (9.15) with $t = r = 1$; then solve for $\hat{\Sigma}_2$ and $\hat{\mathbf{M}}_2$, based on $\mathbf{Z}_2$ and $\mathbf{Z}_1$ and (9.15) and (9.13); then from (9.12) obtain

$$\hat{\mathbf{Q}}_2 \equiv \hat{\Sigma}_2 - \hat{\mathbf{M}}_2 \hat{\Sigma}_1 \hat{\mathbf{M}}_2'; \quad (9.17)$$

then solve for $\hat{\Sigma}_3$ and $\hat{\mathbf{M}}_3$ based on $\mathbf{Z}_3$ and $\mathbf{Z}_2$ and (9.15) and (9.13); and so forth. On occasions, there are difficulties encountered that involve estimates of positive-definite matrices that are *not* positive-definite. Kang, Cressie, and Shi (2010) give a method of “lifting” eigenvalues of such estimates in a way that total variability is preserved. Finally, the method-of-moments estimates, $\hat{\Sigma}_1$, $\hat{\mathbf{M}}_2$, $\hat{\mathbf{Q}}_2$, $\hat{\mathbf{M}}_3$, $\hat{\mathbf{Q}}_3$, $\dots$, are obtained, which together with the semivariogram-based estimates, $\hat{\sigma}_\varepsilon^2$ and $\hat{\sigma}_\xi^2$, result in the method-of-moments estimate, $\hat{\Theta}$.

The last stage of the AOD EHM is to substitute $\hat{\Theta}$ into the expressions for $\hat{Y}_t(s_0)$ and $\sigma_t(s_0)$ and to make the prediction and prediction standard error maps by allowing s_0 to vary over (a fine grid of) D_s . We note that the computations to obtain $\hat{\Theta}$ are extremely fast (Section 9.2.3).

9.2.3 Predictions for Aerosol Optical Depth

The study region D_s we chose is the rectangular area between longitudes 125° W and 3° E and between latitudes 20° S and 44° N, which covers North and

South America, the western part of the Sahara Desert in Africa, the Iberian Peninsula in Europe, and parts of the Atlantic and Pacific Oceans; a map of the region is given in Figure 9.6. Recall that the dataset consists of the MISR Level 3 AOD data collected between July 1 and August 17, and we chose eight days as the time unit; that is, time unit 1 corresponds to July 1–8, time unit 2 corresponds to July 9–16, ..., and time unit 6 corresponds to August 10–17. The number of data in each of these time units is, respectively, $m_1 = 20,970$, $m_2 = 19,398$, $m_3 = 20,819$, $m_4 = 20,167$, $m_5 = 21,759$, and $m_6 = 20,625$. The choice of 8 days allowed enough spatial coverage to estimate the parameters of the EHM. Although the evolutionary time scale of AOD is probably on the order of a day, there is still quite strong temporal dependence between eight-day time periods (Kang, Cressie, and Shi, 2010).

AOD data are highly (right) skewed, even when they are aggregated from Level 2 to Level 3. Hence, we used a logarithmic transformation, as described below. An individual datum in a given 8-day period is obtained by taking a *weighted average* of daily Level 3 $\log(\text{AOD})$ values, where the corresponding weight is proportional to $N_d(s)$, the number of Level 2 observations in the Level 3 pixel s on day d . Henceforth, all analyses are done on this $\log(\text{AOD})$ scale. The transformed data for $t = 1, 2, \dots, 6$ are shown in Figure 9.7.

After considerable exploratory data analysis, we divided the study region D_s into three areas: the oceans, the Americas, and the rest (i.e., the land in the

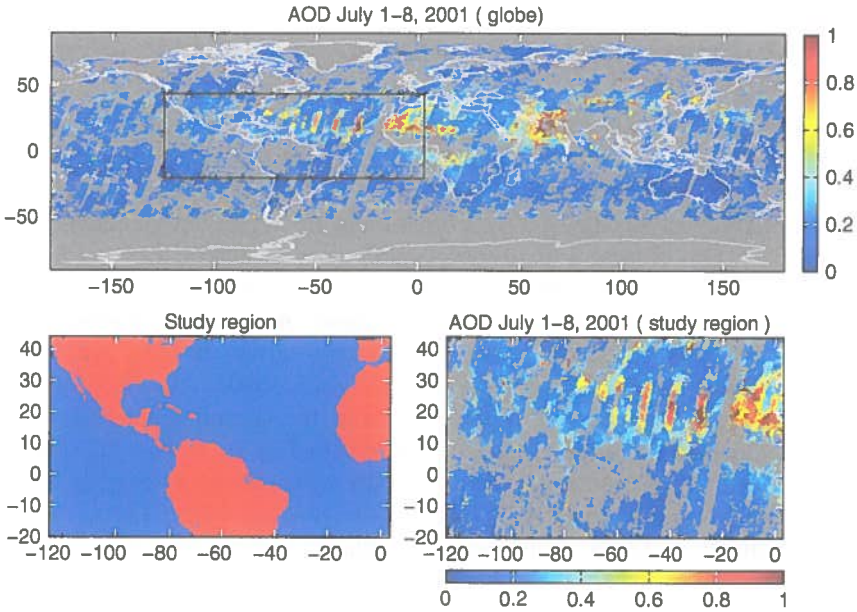


Figure 9.6 Data are retrieved Aerosol Optical Depth (AOD). **Top panel:** Global Level 2 data for $t = 1$ (July 1–8, 2001). **Bottom-left panel:** Study region D_s that includes the Americas and part of Africa (Sahara Desert). **Bottom-right panel:** AOD data in the study region, for $t = 1$.

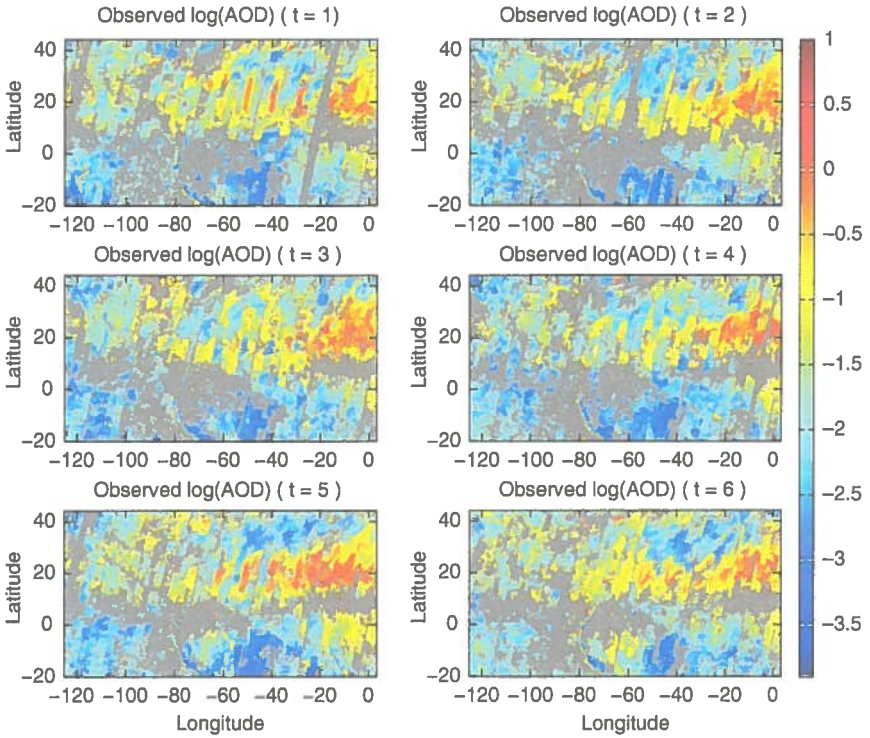


Figure 9.7 Data are (weighted average) log(AOD) at Level 3 ($0.5^\circ \times 0.5^\circ$), for time periods (8 days) $t = 1, 2, \dots, 6$.

eastern part of D_s , which consists of the Sahara Desert and the southwestern tip of Europe). Then, using indicator functions of these regions as explanatory variables, we detrended the original dataset with a spatial-only linear model and ordinary-least-squares (OLS) estimation of the regression parameters from all the data. This created a spatio-temporal *detrended* dataset, $\mathbf{Z}_1, \dots, \mathbf{Z}_6$, upon which the FRF methodology was implemented (assuming mean zero).

Over the period $t = 1, \dots, 6$, we chose $\Phi(\cdot)$ from multiresolution W-wavelet basis functions, using the strategy given by Shi and Cressie (2007), which is based on Huang and Cressie (2000). All 32 W-wavelets from the first scale and 62 W-wavelets from the second scale with “large” absolute coefficients were chosen, which resulted in $p_\alpha = 32 + 62 = 94$, for all t . Recall that Level 3 data are generally obtained by averaging Level 2 data. Hence, we assume in (9.6) that $v_t(\mathbf{s}) = 1/\tilde{N}_t(\mathbf{s})$, where $\{\tilde{N}_t(\mathbf{s}) : \mathbf{s} \in D_s\}$ is the number of Level 2 observations obtained during time unit t in those $0.5^\circ \times 0.5^\circ$ pixels where there were Level 2 data.

Parameter Estimation for the MISR Data

Using the methodology described in Section 9.2.2, we obtained, $\hat{\sigma}_\varepsilon^2 = 0.0191$ and $\hat{\sigma}_\xi^2 = 0.0310$. Figure 9.8 shows the behavior of the robust semivariogram

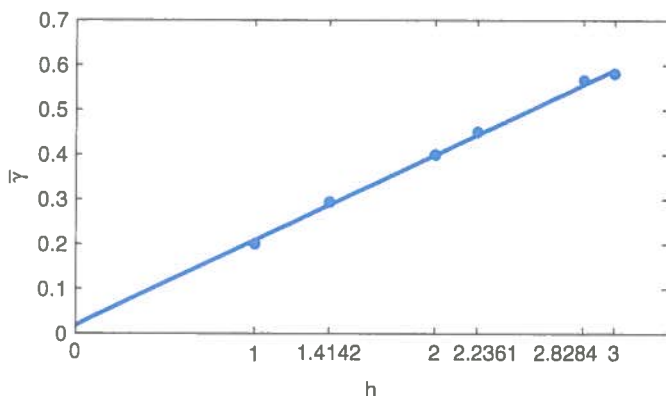


Figure 9.8 Empirical (robust) semivariogram versus spatial lag h , for h near 0. A weighted-least-squares, straight-line fit (Cressie, 1985), $\hat{\gamma}(h)$, is shown. Its intercept at $h = 0$ is $\hat{\sigma}_\varepsilon^2 = 0.0191$.

estimator (based on all data $\mathbf{Z}$) for small spatial lags h , from which $\hat{\sigma}_\varepsilon^2$ was derived.

The method-of-moments estimation of Σ_1 , $\{\mathbf{M}_{t+1}\}$, and $\{\mathbf{Q}_{t+1}\}$ is based on binned empirical covariance and lag-one cross-covariance matrices, as described in Section 9.2.2. The estimate, $\hat{\Sigma}_1$, upon which all other estimates depend, has to be positive-definite. By “lifting” negative eigenvalues (and reducing some positive ones) and ensuring that the sum of the lifted eigenvalues is equal to the sum of the original eigenvalues (Kang, Cressie, and Shi, 2010), it was found that all method-of-moments estimates of the $p_\alpha \times p_\alpha$ matrices were in their respective parameter spaces and, by definition, total variability was preserved.

Upon substituting the estimates $\hat{\sigma}_\varepsilon^2$, $\hat{\sigma}_\xi^2$, $\hat{\Sigma}_1$, $\{\hat{\mathbf{M}}_{t+1}\}$, and $\{\hat{\mathbf{Q}}_{t+1}\}$, into $\hat{Y}_t(s_0)$ given by (9.16) and into $\sigma_t(s_0)$, we obtain AOD EHM predictions of $Y_t(s_0)$ and the predictor’s standard error, respectively. An illustration of the results of these predictions is seen in Figure 9.9. Shown are the observed $\log(\text{AOD})$ values (original data without the trend removed), the OLS trend, the final prediction maps produced by this FRF procedure *with the trend added back*, and the associated standard errors. Notice that the latter map displays the same pattern as seen in the map of the $\log(\text{AOD})$ data, as expected.

Large $\log(\text{AOD})$ values are observed/predicted in the region of the Sahara Desert, decreasing gradually farther to the west. From the prediction map in Figure 9.9, values for the gray regions, where no $\log(\text{AOD})$ data were observed, have been inferred, and the associated standard errors shown in Figure 9.9 give larger values at locations where there were no data, as expected. Finally, the predictions are smoother than the observed data (again, as expected). The presence of the fine-scale variation, $\xi_t(\cdot)$, in the model, is meant to capture a possibly nonsmooth process, and filtering it jointly with α_t represents a departure from the usual filtering methodology (e.g., Section 7.2.1). The fine-scale process allows for “rougher” predictions and more conservative prediction

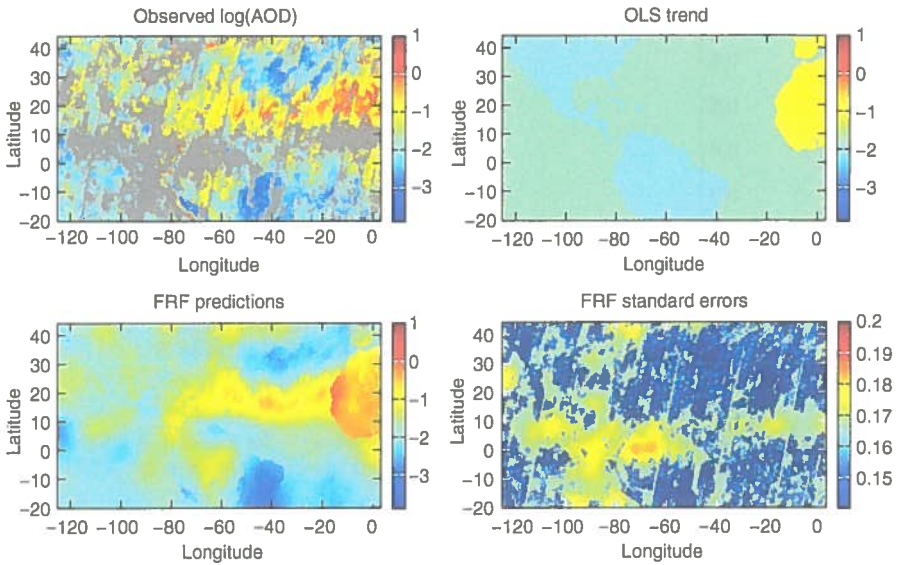


Figure 9.9 Fixed Rank Filtering (FRF) at $t = 6$. **Upper-left panel:** $\log(\text{AOD})$ data at $t = 6$. **Upper-right panel:** Ordinary-least-squares (OLS) fitted trend for all $t = 1, 2, \dots, 6$. **Lower-left panel:** Optimal FRF prediction based on data $\mathbf{Z}_1, \dots, \mathbf{Z}_6$. **Lower-right panel:** FRF standard errors of prediction.

standard errors. Finally, back-transforming the $\log(\text{AOD})$ predictions to the original AOD scale can be done using the results given in Cressie (1993, Section 3.2.2); this must be done if the predictions are to be used as input to climate models.

For this spatio-temporal dataset with $128 \times 256 = 32,768$ pixels and sample size around 120,000, the computation time for binning and fitting the parameters was about a minute on a 2010 vintage workstation. Furthermore, it took about another minute and a half to calculate the FRF predictions and standard errors, up to and including $t = 6$. It is also worth pointing out that the incremental time for updating the Kalman filter at time 6, based on the new data $\mathbf{Z}_6$ and the filtered results at time $t = 5$, was only about 20 seconds. For more details on the computational complexity of this procedure, see Cressie, Shi, and Kang (2010).

We did not give computation time for the other three examples in this chapter, which are based on BHMs and MCMC; implementation can take on the order of minutes to days. While the EHM gives uncertainty measures that are generally too liberal, its computations can be very fast, a factor that is definitely important as spatio-temporal datasets and implementation issues loom large. This is what we referred to as a data/model/computing compromise in Section 2.5.

9.3 MODELING AND FORECASTING THE EURASIAN COLLARED DOVE INVASION

Ecologists have long been interested in the growth and dispersal of biotic organisms. In particular, the dynamics of invasive species are important because the presence of nonnative animals and plants can lead to significant impact on native biotic communities. Using DSTMs in a BHM is a natural framework for modeling invasive species, given the uncertainties in observations, process dynamics, and associated parameters. Wikle (2003b), Wikle and Hooten (2006) and Hooten and Wikle (2007) show how explicit nonlinear PDE dynamics for invasive species can be incorporated into a BHM.

Rather than employing PDE dynamics, Hooten et al. (2007) used a so-called “matrix model” (e.g., Caswell, 2001) for modeling the invasive Eurasian Collared Dove (*Streptopelia decaocto*). We describe their approach in this section. Critically, this example considers the non-Gaussian nature of the observations and the inherent uncertainty associated with the probability of detection. The dynamics are nonlinear (indeed, they are a special case of the generalized quadratic nonlinearity described in Section 7.3.3), and they are considered within a population model that includes density-dependent growth and dispersal. Another feature of this example is that it illustrates the importance of allowing the process-model parameters (e.g., dispersal rates) to vary spatially.

9.3.1 The Eurasian Collared Dove Invasion of North America

The Eurasian Collared Dove (ECD) was first observed in North America in the early 1980s (Romagosa and Labisky, 2000). Just as they spread through Europe in the early 1900s, they are spreading quickly through North America, and they are posing a potential threat to native ecosystems (e.g., Hengeveld, 1993).

As part of the North American Breeding Bird Survey (BBS; Robbins, Bystrak, and Geissler, 1986), each year, during the peak of the avian breeding season, skilled observers collect bird-population data along predetermined roadside survey routes. Each survey route is approximately 39.4 km in length, and an observer makes 50 stops over each route. At each stop, the observer counts and identifies, by sight or sound for a 3-minute period, all bird species observed within an approximate 0.4-km radius. Not surprisingly, this presents a fairly complicated spatial unit of measurement. In this example, we assume that the data collected along each route corresponds to a count at the route center. Given that our interest is on the invasion at a continental scale, the effects of such aggregation should be quite minimal; however, we note that this aggregation does influence how we should interpret model parameters. We also note that BBS data are subject to other sources of uncertainty (e.g., Link et al., 1994; Sauer, Peterjohn, and Link, 1994), which suggests that we should try to account for these data uncertainties in our BHM.

Thus, (9.18) is the data model used in the ECD BHM. Importantly, it accommodates the uncertainty associated with the imprecise sampling of the ECD, by making use of previously published empirical information to inform the data-model parameters.

9.3.3 ECD Process Model

In order to perform ecologically meaningful inference (spatial prediction and forecasting), a model must be specified for the dynamical process that is based on known characteristics of invasive species. Although it is possible to perform transformations on the non-Gaussian state process (which is a count process in this case) to simplify modeling, it can sometimes be difficult to incorporate scientifically meaningful information into a model for the transformed state (or, to extract such information, once the model has been fitted). Thus, in this analysis we consider a more intuitive construction of the process model that seeks to retain ecologically meaningful dynamical behavior.

Recall that the true ECD population (i.e., abundance) at time t and spatial location $\mathbf{s}_i$ is represented by $Y_t(\mathbf{s}_i)$. We assume that, conditional on the true intensity, this follows an independent Poisson distribution. Furthermore, we define the m -dimensional vector (corresponding to observation locations), $\mathbf{Y}_t \equiv (Y_t(\mathbf{s}_1), \dots, Y_t(\mathbf{s}_m))'$, and we assume that,

$$\mathbf{Y}_t | \lambda_t \sim \text{ind. Poi}(\mathbf{H}\lambda_t), \quad t = 1, 2, \dots,$$

where the n -dimensional vector, $\lambda_t \equiv (\lambda_t(\mathbf{s}_1), \dots, \lambda_t(\mathbf{s}_m), \lambda_t(\mathbf{s}_{m+1}), \dots, \lambda_t(\mathbf{s}_n))'$ corresponds to the observation locations *and* the additional predictor locations $\{\mathbf{s}_{m+1}, \dots, \mathbf{s}_n\}$. The matrix $\mathbf{H}$ is an incidence matrix (see Section 7.1.2) that relates the true process $Y_t(\cdot)$ at observation locations, to the intensity process λ_t at all locations of interest. Now, we assume that the Poisson intensities evolve according to a quadratically nonlinear DSTM (Section 7.3.3). In particular, we use a “matrix model” framework from population dynamics (Caswell, 2001, Chapter 3), but with *random* parameters. Specifically, consider the following process model for the ECD intensities

$$\begin{aligned} \lambda_t &= \mathbf{M}\lambda_{t-1} \\ &= \mathbf{B}(\boldsymbol{\tau})\mathbf{G}(\lambda_{t-1}; \boldsymbol{\theta}^G)\lambda_{t-1}, \quad t = 2, 3, \dots, \end{aligned} \quad (9.19)$$

where the propagator matrix $\mathbf{M}$ is comprised of two distinct $n \times n$ matrices. Specifically, $\mathbf{G}(\lambda_{t-1}; \boldsymbol{\theta}^G)$ is a diagonal matrix that accommodates growth over time and is dependent on the previous state λ_{t-1} and growth parameters $\boldsymbol{\theta}^G$; and $\mathbf{B}(\boldsymbol{\tau})$ accommodates dispersal of the population and is dependent on dispersal parameters $\boldsymbol{\tau}$. In this case, the initial condition λ_1 must also be specified.

Notice the distinct lack of an error term in (9.19). Although this model is deterministic when conditioned on the parameters and initial state, it is random when the random nature of these quantities is considered. Additive error processes can be considered for latent invasive-species dynamics (e.g.,

Wikle, 2003b; Hooten and Wikle, 2007), but they can complicate the analysis computationally, as well as inferentially (in terms of parameter interpretation and the lack of “smoothness” in the forecasts). Hooten et al. (2007) show that allowing the initial state and parameters to be random alleviates these complications yet provides flexibility for the latent process to adapt to the data. One convenient feature of this parameterization is that, given an initial condition for a set of prediction locations (that may contain the observation locations), say λ_1 , and the corresponding model parameters, θ^G and τ , one immediately obtains the predictions λ_t for any time t , by iterating the model (9.19) forward in time. This allows $\lambda_t(\cdot)$ (and, eventually, $Y_t(\cdot)$) to be predicted easily and to be forecasted at any spatial location of interest.

Here, we parameterize the diagonal matrix $\mathbf{G}$ in terms of a Ricker growth equation (e.g., see Turchin, 2003, Chapter 3). That is, the i th diagonal element of $\mathbf{G}$ is parameterized as follows:

$$G_{ii}(\theta_1^G, \theta_2^G, \lambda_t(s_i)) \equiv \exp \left\{ \theta_1^G \left(1 - \frac{\lambda_t(s_i)}{\theta_2^G} \right) \right\}, \quad i = 1, \dots, n,$$

where θ_1^G and θ_2^G are the growth and carrying-capacity parameters, respectively. One could certainly imagine scenarios where one or both of these parameters could vary across space. However, it is difficult to estimate carrying capacities for populations that are still growing, and thus a spatially explicit carrying-capacity parameter would have to rely heavily on prior information. Similarly, in the presence of spatially varying dispersal, estimation of spatially varying growth rates for such a transient population would need strong prior information. In the absence of such prior information, we choose to focus on the spatially varying dispersal term (see below) and do not allow the growth and carrying capacities to vary spatially.

The dispersal in (9.19) is governed by the matrix $\mathbf{B}(\tau)$. Specifically, analogous to the IDE models presented in Section 7.2.5, we use a Gaussian kernel to characterize dispersal, such that the population at a specific location at time t is a weighted average of the population at surrounding locations at time $t - 1$. In this case, we suppose that the (i, j) th element of the $n \times n$ matrix $\mathbf{B}(\tau)$ is given by

$$B_{i,j}(\tau) \propto \exp \left\{ -\frac{d_{i,j}^2}{\tau(s_i)} \right\},$$

where $d_{i,j}$ is the distance between location s_i and s_j , and $\tau \equiv (\tau(s_1), \dots, \tau(s_n))'$ are spatially varying dispersal coefficients. In this formulation, it is assumed that each row of $\mathbf{B}(\tau)$ is constrained to sum to one. As noted by Hooten et al. (2007), other parameterizations are possible, but this particular parameterization corresponds to a discretized diffusion equation with a population that can pass through “open boundaries,” something that is typical of invasive species.

9.3.4 ECD Parameter Model

Since we are interested in predicting at locations different from our observations, we specify a spatial model for λ_1 and τ . In both cases, we assume that the spatial processes are log-Gaussian with an exponential covariance function (see Section 4.1.1). In addition, the population-growth parameter, θ_1^G , and the carrying-capacity parameter, θ_2^G , are assumed to be independent and follow a Gaussian distribution and an inverse gamma distribution, respectively. As detailed in Hooten et al. (2007), considerable care went into the choice of the hyperparameters for these parameter models, as well as the hyperparameters for the spatial models associated with the initial condition and dispersal coefficients. Finally, recall from Section 9.3.2 that empirical results from Link et al. (1994) were used to specify the data-model parameters, a_θ and b_θ .

9.3.5 The ECD BHM: Implementation and Inference

The implementation details of the ECD BHM can be found in Hooten et al. (2007) and the associated online appendices cited therein. In summary, prediction and forecasting was implemented on a grid of points across the eastern two-thirds of the United States. The MCMC algorithm, which was a combination of Gibbs and Metropolis–Hastings steps, was run for 200,000 iterations, the first 20,000 of which were set aside for “burn-in.”

The primary focus in this example is on spatial prediction and forecasting of ECD population abundance, $Y_t(\cdot)$, at various spatial locations in the United States. Summaries of these predictions can be shown as spatial maps. Figure 9.11 shows the maps of the posterior mean of the ECD abundance for each year from 1986 through 2003. Note that, although these maps are displayed as images, where the color of the grid cell is representative of the ECD abundance, the prediction actually corresponds to a *point* in the middle of the box. The image format is for clarity of presentation and is not meant to imply that prediction is for an areal spatial support. Note also that the white areas of the map were not considered in the analysis. These correspond to the ocean, Mexico, and the northeast United States, where there were no ECD data available during the 1986–2003 time period. This figure gives a large-scale perspective of the ECD invasion that was in progress through 2003. Additionally, Figure 9.12 shows the posterior mean of ECD-abundance forecasts through the year 2020. This figure suggests that the ECD is on track to successfully colonize this entire region of the United States by about 2016 (i.e., it is forecasted to be at, or near, its carrying capacity by this time).

Obviously, when making predictions or forecasts, one should examine the associated prediction and forecast uncertainties. The prediction and forecast uncertainty maps for ECD abundance look very much like the posterior mean maps shown in Figures 9.11 and 9.12; this is not surprising, given that we are assuming that the abundance follows a (conditional) Poisson distribution (for which the mean is equal to the variance). It is more informative to examine these distributions in the context of time series of abundance predictions for

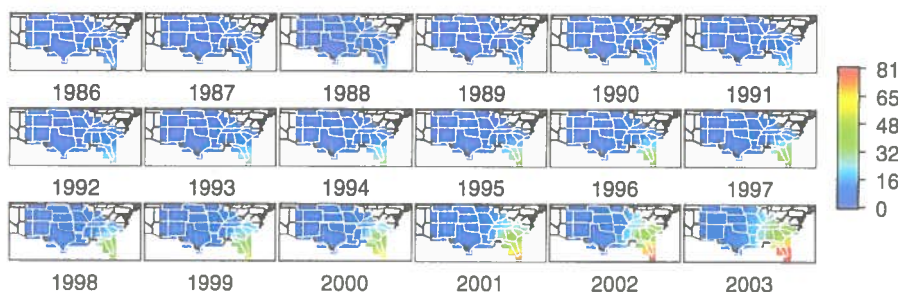


Figure 9.11 Posterior means of ECD abundance in the United States for the period 1986 to 2003 (corresponding to the sampling period considered in this analysis). Areas of the map shown as “white” were not included in the prediction grid.

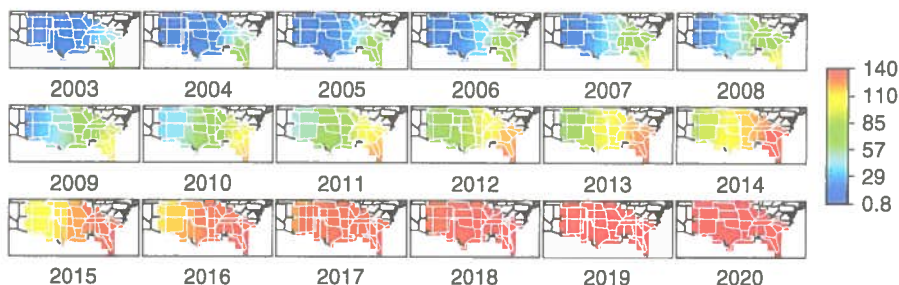


Figure 9.12 Posterior mean of the ECD-abundance forecast in the United States for the period 2003 to 2020 (where the years 2004–2020 correspond to out-of-sample forecasts). Areas of the map shown as “white” were not included in the prediction grid.

specific locations. For example, Figure 9.13 shows the posterior 95% credible intervals for ECD abundance at a location in south Florida, one in eastern Kansas, and one in northern Utah. Clearly, the population at the south Florida location likely reached its carrying capacity in the early 2000s, whereas the Kansas and Utah locations are forecasted to reach their carrying capacities in the middle and late 2010s, respectively.

Although the main goal for this analysis was to spatially predict and forecast abundance, gaining knowledge about the underlying process parameters that govern the population dynamics can also be useful. As described in Hooten et al. (2007), the posterior distributions for the growth parameters were substantially narrower than their prior distributions, which suggested that the data were informative for inference on these parameters. Furthermore, the posterior distribution associated with the spatially varying dispersal parameters also suggested “learning.” Figure 9.14 shows the posterior mean of the dispersal parameters on the grid used for spatial prediction and forecasting. Note that the magnitudes of these dispersal parameters are directly related to the scale of the map-distance units; therefore, it is the overall pattern and relative magnitude that

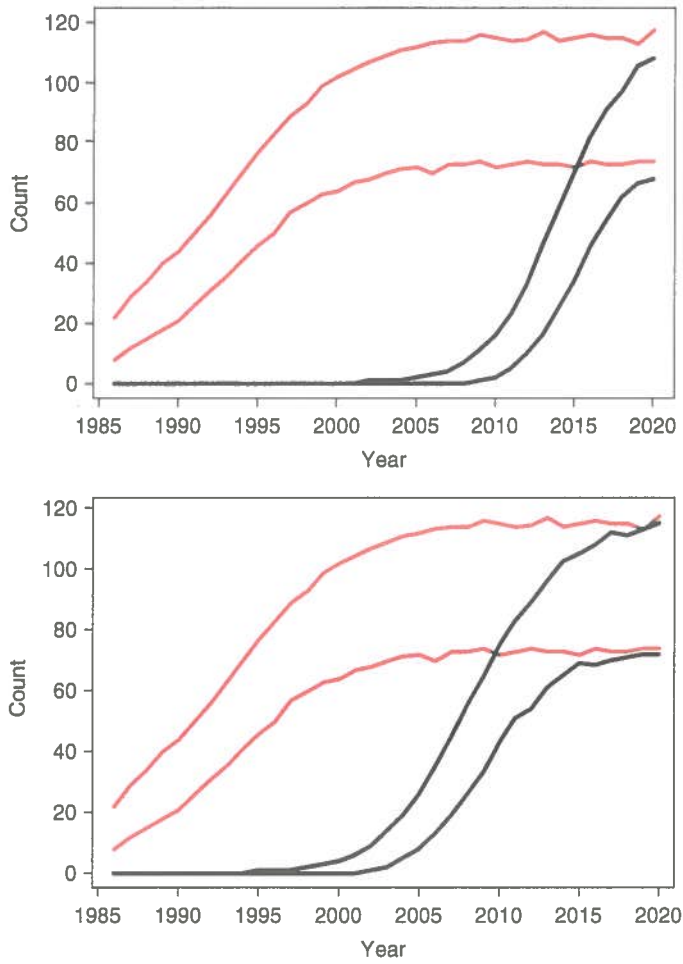


Figure 9.13 Comparisons of posterior credible intervals for the ECD population from 1986 to 2020. **Top panel:** Credible intervals for a location in south Florida (red) and for northern Utah (black). **Bottom panel:** Credible intervals for the south Florida location (red) and for eastern Kansas (black).

is important, not the actual value of the estimate. This map shows that there is clearly an area of lower dispersal in northern Florida, which would suggest that the ECD invasion was relatively slower to spread north and westward out of Florida. In addition, the dispersal in the western United States is likely to be faster than in the southeast. Hooten et al. (2007) show, using the deviance information criterion (DIC) measure (e.g., Spiegelhalter et al., 2002), that the model with spatially varying (heterogeneous) dispersal coefficients fits the data better than the model assuming constant (homogenous) dispersal.

The analysis shown here was originally performed in the mid-2000s, based on data through 2003. It is interesting to examine the state of the ECD invasion

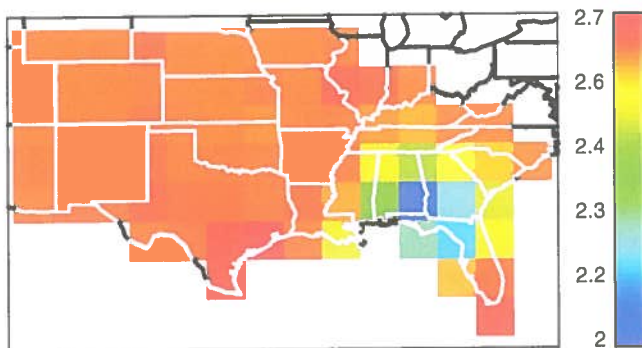


Figure 9.14 Posterior mean of the spatially varying dispersal coefficients (τ) in the ECD process model (9.19).

of North America in 2010. Maps of user-reported ECD sightings through the Spring of 2010 show relatively frequent ECD encounters throughout the continental United States, with the exception of the northeast (e.g., see the maps of ECD “checklist frequency” from the eBird project; eBird, 2010). Thus, the forecast shown in Figure 9.12 was quite reasonable, if somewhat conservative, in the speed of the invasion. Presumably, the forecast could have been improved if ecologically relevant covariates were used in the model for the dispersal parameters.

In summary, this approach for modeling invasive species was able to take into account uncertainty in the detection of ECDs in the North American BBS observations, within the context of a scientifically intuitive and useful nonlinear DSTM that could be used in a BHM to predict and forecast the ongoing ECD invasion. This illustrates the power of the BHM approach for combining observations and scientific information in the presence of uncertainty.

9.4 MEDITERRANEAN SURFACE VECTOR WINDS

Beginning in the late 1990s, a sequence of increasingly sophisticated BHMs for near-surface wind fields (so-called “surface vector winds,” or SVWs) were developed using satellite scatterometer observations. In particular, Royle et al (1999) derived a BHM for SVWs based on a single snapshot in time (i.e. a spatial model); data were made up of two swaths of satellite scatterometer SVWs in the Labrador Sea region. The process model in Royle et al. (1999) was based on a PDE representing a *geostrophic-ageostrophic* partition (e.g., Holton 2004, Chapter 3) of the SVW field. Sea-level pressure (SLP) was represented by a latent spatial process model with random parameters. As described in Section 5.5, conditioning multivariate processes on a common process is a very effective approach for building dependence, especially if, as in this case there is strong scientific motivation for the conditioning.

Similarly, Wikle et al. (2001) developed an SVW BHM for tropical wind using satellite scatterometer data, in addition to weather-center reanalysis winds. The process-model component of their BHM is also based on substantial scientific knowledge, both from theory and empirical study. In particular, the process is made up of three components: (1) a spatial mean component accounting for land–sea contrasts, (2) the normal modes (Section 5.5) of the equatorial shallow water equations (at large scales), and (3) a turbulence-closure model based on observed kinetic energy spectra for the tropical surface winds (e.g., Wikle, Berliner, and Cressie, 1998). Implementation of the closure-model component used nested wavelet bases (i.e., multiresolution analysis). Critically, the normal mode and wavelet coefficients were assumed to evolve according to a linear dynamical model with hyperparameters suggested by independent empirical results from the literature (Wheeler and Kiladis, 1999). Realistic surface convergence features in the posterior wind distributions were shown to correspond well with intense atmospheric deep convection and rainband signatures in infrared cloud imagery during the mature phase of tropical cyclone Dale (Wikle et al., 2001). Hoar et al. (2003) extended this model to generate a multi-year dataset of 50 realizations from the SVW posterior distribution for the tropical Indian and western Pacific oceans. Mapes, Milliff, and Morzel (2009) combined the winds reported by Hoar et al. (2003) with cloud and precipitable-water datasets to identify lead-lag relationships in the life cycles of over 10,000 mesoscale convective systems (MCS) in the tropical Indian and western Pacific Oceans.

Berliner, Milliff, and Wikle (2003) demonstrated how a BHM could handle multi-platform observations in the data model, where the process model is based on an idealized coupled air–sea interaction model in a high-latitude, partially enclosed sea. A key feature of this study is the demonstration that a process model based on discretized quasi-geostrophic dynamics (see Section 7.4; Wikle and Hooten, 2010), with random parameters and additive error, is sufficient to fully capture the nonlinear dynamics from a simulated truth based on shallow-water equations. This is so, even with relatively sparse observations.

The latest SVW BHM in this series of models is due to Milliff et al. (2011), who extend the geostrophic–ageostrophic process model for the SVW to include explicit surface-friction effects. Data include satellite retrievals of SVW, as well as SVW and sea-level-pressure fields from weather-center analyses. The energy spectrum of the weather-center winds is not realistic at the spatial scales of interest here, but the scatterometer data do represent realistic energy content. Thus, one of the goals of this analysis was to merge systematically these two datasets, in order to obtain posterior realizations that have a realistic energy spectrum.

We focus on an extension of the model and analysis of Milliff et al. (2011) here, as discussed in detail in Bonazzi (2008). In particular, this example shows how a science-based framework can suggest a *conditional* multivariate model in the presence of multiple data sources. More importantly, the broader

goal of this study is to provide SVW forcing fields for an operational deterministic, ocean-forecast model for the Mediterranean Sea. These forcing winds have realistic spatio-temporal structure (in terms of means, variances covariances, and energy spectra) relative to known regional wind regimes. Such realistic variation in the SVW leads to realistic uncertainties in the ocean fields used to initialize the forecast model and, thus, it leads to more realistic assessments of forecast accuracy (Milliff et al., 2011; Pinardi et al., 2011).

Before we describe the SVW BHM, we give a little background on how an ocean-forecast system works. In particular, we consider the Mediterranean Forecast System (MFS), which produces 10-day forecasts of the ocean state variables (e.g., three-dimensional fields of temperature, pressure, density, salinity, flow vectors, sea-surface height, etc.) every day (for details, see Pinardi et al., 2003; Dobricic et al., 2005; Pinardi et al., 2011). The forecast system is comprised of a data-assimilation part leading to a forecast of initial ocean conditions, along with a deterministic ocean-forecast model. The underlying ocean model in this system represents a discretization (in three dimensions) of the PDEs that govern ocean dynamics and thermodynamics. In the MFS, a *sequential* data assimilation method is used, in which the ocean state variables are adjusted daily from a first guess (prior forecast) by incorporating multivariate observations of the Mediterranean Sea. These observations accumulate over a 14-day period leading up to the initial time of the forecast. In this way, the forecast model launches from an initial state that includes the influence of the latest observations. During the data-assimilation step, the sequential procedure is forced by observed winds and, during the forecast step, the deterministic ocean-forecast-model integration is forced by forecasted SVW from a fitted BHM. One way to obtain uncertainty measures for such forecasts is to start with multiple initial conditions (so-called “ensembles”) and generate an ensemble of forecasts. It is this component of the system, namely generating ensembles, that the SVW BHM is designed to address.

9.4.1 Scientific Motivation for the Process Model

The process model for SVW is motivated by an approximation to the full momentum and thermodynamic equations for the surface wind, known as the *Rayleigh friction equations* (RFE; e.g., Stevens et al., 2002):

$$\begin{aligned}\frac{\partial u}{\partial t} - f v &= -\frac{1}{\rho_0} \frac{\partial p}{\partial x} - \gamma u, \\ \frac{\partial v}{\partial t} + f u &= -\frac{1}{\rho_0} \frac{\partial p}{\partial y} - \gamma v,\end{aligned}\tag{9.20}$$

where u and v are the east–west and north–south components of the surface winds (and are functions of time and space), respectively; f is the Coriolis parameter; ρ_0 is the reference atmospheric density; p is the sea-level pressure (also a function of time and space); and γ is the Rayleigh friction term. These

equations are important in that they allow friction effects to be captured (which are important for surface winds, even over the ocean) with linear dynamics. Following an approach similar to Milliff et al. (2011), one can show that a reasonable low-order approximation to the RFE equations leads to the following representation of the velocity components (e.g., Bonazzai, 2008)

$$\begin{aligned}
 u &\simeq -\frac{f}{\rho_0(f^2 + \gamma^2)} \frac{\partial p}{\partial y} - \frac{\gamma}{\rho_0(f^2 + \gamma^2)} \frac{\partial p}{\partial x} \\
 &\quad - 2 \frac{\gamma}{\rho_0(f^2 + \gamma^2)} \frac{\partial u}{\partial t} - \frac{1}{\rho_0(f^2 + \gamma^2)} \frac{\partial^2 p}{\partial x \partial t}, \\
 v &\simeq \frac{f}{\rho_0(f^2 + \gamma^2)} \frac{\partial p}{\partial x} - \frac{\gamma}{\rho_0(f^2 + \gamma^2)} \frac{\partial p}{\partial y} \\
 &\quad - 2 \frac{\gamma}{\rho_0(f^2 + \gamma^2)} \frac{\partial v}{\partial t} - \frac{1}{\rho_0(f^2 + \gamma^2)} \frac{\partial^2 p}{\partial y \partial t}.
 \end{aligned} \tag{9.21}$$

If we consider centered differences to replace the spatial derivatives in (9.21), we can write a process model in vector form as

$$\begin{aligned}
 \mathbf{u}_t &= a_1^{(u)} \mathbf{D}_y \mathbf{p}_t + a_2^{(u)} \mathbf{D}_x \mathbf{p}_t + a_3^{(u)} \mathbf{u}_{t-1} + a_4^{(u)} \mathbf{D}_x \mathbf{p}_{t-1} + \boldsymbol{\eta}_t^{(u)}, \\
 \mathbf{v}_t &= a_1^{(v)} \mathbf{D}_x \mathbf{p}_t + a_2^{(v)} \mathbf{D}_y \mathbf{p}_t + a_3^{(v)} \mathbf{v}_{t-1} + a_4^{(v)} \mathbf{D}_y \mathbf{p}_{t-1} + \boldsymbol{\eta}_t^{(v)},
 \end{aligned} \tag{9.22}$$

where $\mathbf{u}_t$ and $\mathbf{v}_t$ represent a vectorization of the gridded wind components; $\mathbf{p}_t$ and $\mathbf{p}_{t-1}$ represent vectorizations of the gridded pressure field at time t and $t - 1$, respectively; $\{a_i^{(\cdot)} : i = 1, \dots, 4\}$ are parameters corresponding to the respective coefficients in front of the partial derivative terms in (9.21); $\mathbf{D}_x$ and $\mathbf{D}_y$ are matrix operators that give the x -direction and y -direction centered differences, respectively; and $\boldsymbol{\eta}_t^{(u)}$ and $\boldsymbol{\eta}_t^{(v)}$ are additive error terms that represent the effects of the approximation used to obtain (9.21). These error processes are further decomposed in terms of wavelet basis functions, as described in Section 9.4.2.

9.4.2 The SVW BHM

As is customary with BHM implementations, we distinguish between the data-model, process-model, and parameter-model levels of the hierarchy.

Data Model

In this application, we consider two sources of wind data and one source of sea-level pressure data. In particular, we have satellite wind observations from the QuikSCAT scatterometer, along with surface winds and pressures from

an analysis by the European Center for Medium range Weather Forecasting (ECMWF). The specific data models are

$$\begin{aligned} \mathbf{d}_t^{Q_u} | \mathbf{u}_t, \sigma_Q^2 &\sim \text{ind. Gau}(\mathbf{H}_t^Q \mathbf{u}_t, \sigma_Q^2 \mathbf{I}), \\ \mathbf{d}_t^{Q_v} | \mathbf{v}_t, \sigma_Q^2 &\sim \text{ind. Gau}(\mathbf{H}_t^Q \mathbf{v}_t, \sigma_Q^2 \mathbf{I}), \\ \mathbf{d}_t^{E_u} | \mathbf{u}_t, \sigma_E^2 &\sim \text{ind. Gau}(\mathbf{H}_t^E \mathbf{u}_t, \sigma_E^2 \mathbf{I}), \\ \mathbf{d}_t^{E_v} | \mathbf{v}_t, \sigma_E^2 &\sim \text{ind. Gau}(\mathbf{H}_t^E \mathbf{v}_t, \sigma_E^2 \mathbf{I}), \\ \mathbf{d}_t^{E_p} | \mathbf{p}_t, \sigma_{E_p}^2 &\sim \text{ind. Gau}(\mathbf{H}_t^E \mathbf{p}_t, \sigma_{E_p}^2 \mathbf{I}), \end{aligned}$$

where $\mathbf{d}_t^{Q_u}$ and $\mathbf{d}_t^{Q_v}$ are vectors of scatterometer u -wind and v -wind observations, respectively, within a specified time window indexed by t ; and $\mathbf{d}_t^{E_u}$, $\mathbf{d}_t^{E_v}$, and $\mathbf{d}_t^{E_p}$ are ECMWF u -wind and v -wind component and pressure observations, respectively, within the same time window. In addition, recall that $\mathbf{u}_t$, $\mathbf{v}_t$, and $\mathbf{p}_t$ represent vectorizations of the gridded wind components and pressure, respectively, of the true processes. The mapping matrices (incidence matrices in this case; see Section 7.1.2) for the scatterometer and ECMWF observations are given by $\mathbf{H}_t^Q$ and $\mathbf{H}_t^E$, respectively. Finally, measurement errors are assumed to be independent in space and time, conditional upon the true process values; they have Gaussian distributions with mean zero and measurement-error variances, σ_Q^2 , σ_E^2 , and $\sigma_{E_p}^2$ (corresponding to scatterometer wind components, ECMWF wind components, and ECMWF pressures, respectively).

A sample of the wind data for February 2, 2005 is shown in Figure 9.15. These panels show the QuikSCAT scatterometer and ECMWF analysis observations available within a window of ± 3 h of $t = 00:00, 06:00, 12:00$, and $18:00$ UTC (“Coordinated Universal Time”). Note that the ECMWF-analysis winds and pressures are available on a $0.5^\circ \times 0.5^\circ$ spatial grid, and they are available at each time period for all locations. This grid is also used for the process vectors. In contrast, the QuikSCAT observations are available intermittently in space due to the polar orbit of the satellite, but at much higher spatial resolution (25 km) when they are available. In this sense, the mapping matrices for the scatterometer data $\mathbf{H}_t^Q$ are incidence matrices based on the assumption that all scatterometer observations within 0.25° of a process grid-point, and within 3 hours of time t , are associated with the wind process at that gridpoint and time.

Process Model

The process model given by (9.22) is motivated by the RFE. The model can be rewritten in terms of the process distributions:

$$\begin{aligned} \mathbf{u}_t | \mathbf{p}_t, \beta_t^{(u)}, \{a_i^{(u)} : i = 1, \dots, 4\}, \sigma_u^2 &\sim \text{ind. Gau}(\mu_t^{(u)}, \sigma_u^2 \mathbf{I}), \\ \mathbf{v}_t | \mathbf{p}_t, \beta_t^{(v)}, \{a_i^{(v)} : i = 1, \dots, 4\}, \sigma_v^2 &\sim \text{ind. Gau}(\mu_t^{(v)}, \sigma_v^2 \mathbf{I}), \end{aligned}$$

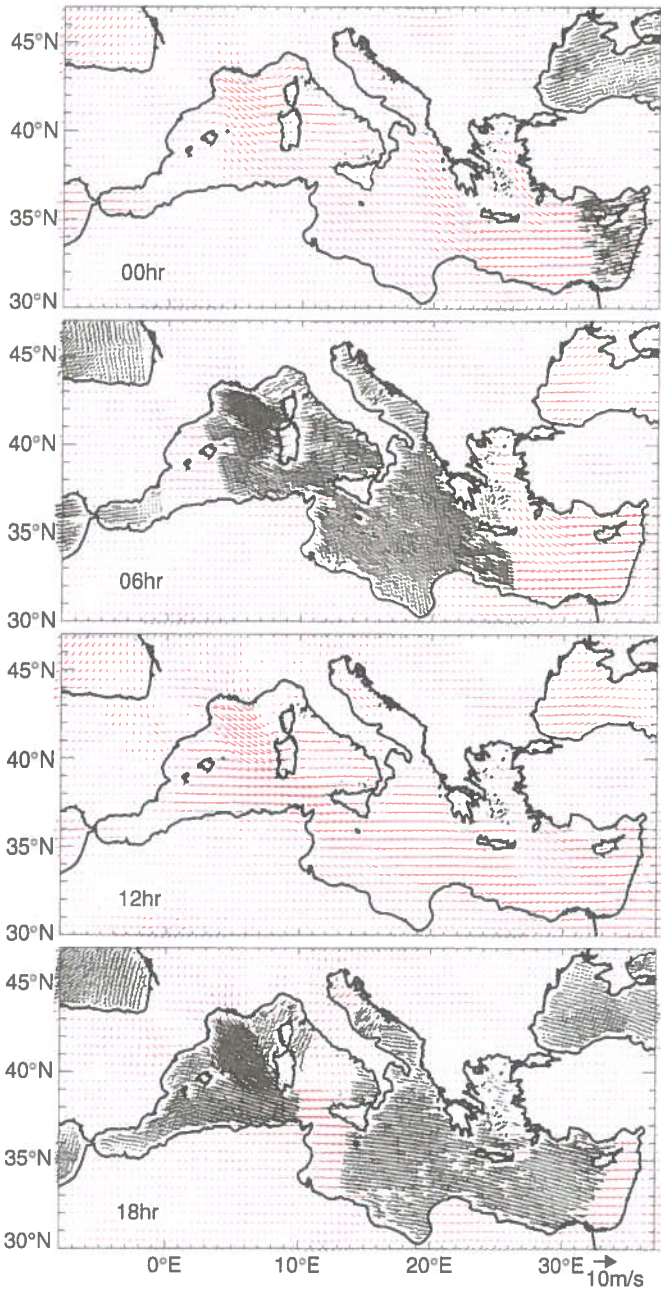


Figure 9.15 Wind observations from February 1, 2005. From top to bottom, the panels correspond to available data at 00:00, 06:00, 12:00, and 18:00 UTC. The red vectors over the Mediterranean Sea correspond to ECMWF-analysis winds on a $0.5^\circ \times 0.5^\circ$ grid. The black vectors correspond to the much-higher-resolution (25 km), but spatially intermittent, wind retrievals from the QuickSCAT scatterometer.

where

$$\begin{aligned}\mu_t^{(u)} &= a_1^{(u)} \mathbf{D}_y \mathbf{p}_t + a_2^{(u)} \mathbf{D}_x \mathbf{p}_t + a_3^{(u)} \mathbf{u}_{t-1} + a_4^{(u)} \mathbf{D}_x \mathbf{p}_{t-1} + \mathcal{W} \boldsymbol{\beta}_t^{(u)}, \\ \mu_t^{(v)} &= a_1^{(v)} \mathbf{D}_x \mathbf{p}_t + a_2^{(v)} \mathbf{D}_y \mathbf{p}_t + a_3^{(v)} \mathbf{v}_{t-1} + a_4^{(v)} \mathbf{D}_y \mathbf{p}_{t-1} + \mathcal{W} \boldsymbol{\beta}_t^{(v)},\end{aligned}$$

and we have assumed that the error processes, $\eta_t^{(u)}$ and $\eta_t^{(v)}$, are decomposed in terms of a wavelet-basis expansion plus white noise. That is, given $\boldsymbol{\beta}_t^{(u)}$ and $\boldsymbol{\beta}_t^{(v)}$, we have

$$\begin{aligned}\eta_t^{(u)} &= \mathcal{W} \boldsymbol{\beta}_t^{(u)} + \mathbf{v}_t^{(u)}; \quad \mathbf{v}_t^{(u)} \sim \text{ind. Gau}(\mathbf{0}, \sigma_u^2 \mathbf{I}), \\ \eta_t^{(v)} &= \mathcal{W} \boldsymbol{\beta}_t^{(v)} + \mathbf{v}_t^{(v)}; \quad \mathbf{v}_t^{(v)} \sim \text{ind. Gau}(\mathbf{0}, \sigma_v^2 \mathbf{I}),\end{aligned}$$

where $\mathcal{W}$ is a matrix of wavelet basis functions, and $\boldsymbol{\beta}_t^{(u)}$ and $\boldsymbol{\beta}_t^{(v)}$ are the associated wavelet coefficients. This formulation is adapted from Wikle et al. (2001) to account for multiscale spatially correlated error processes that have been shown to be important after large-scale features were removed from the scatterometer data [see Milliff et al. (2011) for details].

Now, as in Milliff et al. (2011), we write the vector of gridded pressures as

$$\mathbf{p}_t = \mu_p \mathbf{1} + \Phi \boldsymbol{\alpha}_t,$$

where μ_p represents the overall (known) mean sea-level pressure, Φ is defined as a set of p_α EOF basis functions (where the EOFs were calculated based on historical sea-level pressure from the ECMWF reanalysis), and $\boldsymbol{\alpha}_t$ are the EOF spectral coefficients. Because the pressure field is deterministically related to the spectral coefficients (since the mean μ_p and the basis functions Φ are assumed known), the pressure “process” is determined by $\boldsymbol{\alpha}_t$ on a reduced-dimensional space. Given that we have “complete” data fields for pressure data, a fairly simple model for $\boldsymbol{\alpha}_t$ can be specified:

$$\boldsymbol{\alpha}_t | \boldsymbol{\lambda} \sim \text{iid Gau}(\mathbf{0}, \text{diag}(\boldsymbol{\lambda})),$$

where $\boldsymbol{\lambda}$ is a p_α -dimensional vector with positive entries.

In conclusion, the dynamics in the process model for SVW are not complicated, because the large-scale dynamics are already accounted for in the pressure data. This is an example of a model/data trade-off, in which a less complicated dynamical model is offset by fairly complete observations. It is the wavelet spectral coefficients, $\boldsymbol{\beta}_t^{(u)}$ and $\boldsymbol{\beta}_t^{(v)}$, that are dynamical; it is seen in the parameter models below that they evolve as univariate AR(1) processes.

Parameter Model

The process-model parameters associated with the pressure derivatives are given in (9.22). Their prior distributions are assumed to be

$$a_j^{(u)} \sim \text{Gau}(\tilde{\mu}_j^{(u)}, \sigma_{u,j}^2), \quad a_j^{(v)} \sim \text{Gau}(\tilde{\mu}_j^{(v)}, \sigma_{v,j}^2), \quad j = 1, \dots, 4,$$

independently. Now, the theoretical relationship in (9.21) can be used to suggest the prior means:

$$\begin{aligned} \tilde{\mu}_1^{(u)} &= -\tilde{\mu}_1^{(v)} = \frac{-f}{\rho_0(f^2 + \gamma^2 + 2\gamma/\Delta)} \\ \tilde{\mu}_2^{(u)} &= \tilde{\mu}_2^{(v)} = \frac{-(1/\Delta + \gamma)}{\rho_0(f^2 + \gamma^2 + 2\gamma/\Delta)} \\ \tilde{\mu}_3^{(u)} &= \tilde{\mu}_3^{(v)} = \frac{2\gamma}{\Delta(f^2 + \gamma^2 + 2\gamma/\Delta)} \\ \tilde{\mu}_4^{(u)} &= \tilde{\mu}_4^{(v)} = \frac{1}{\rho_0\Delta(f^2 + \gamma^2 + 2\gamma/\Delta)}, \end{aligned}$$

where f , ρ_0 , and γ take values that are physically realistic for the region under consideration and Δ is the time discretization constant [for details, see Milliff et al. (2011) and Bonazzi (2008)]. Furthermore, the wavelet spectral coefficients are dynamical and are assumed to evolve as univariate AR(1) processes. That is,

$$\beta_t^{(k)}(i) \sim \text{Gau}(m_k(i)\beta_{t-1}^{(k)}(i), \sigma_{\beta_k}^2(i)), \quad \text{for } k = u, v \text{ and } i = 1, \dots, n_\beta,$$

where n_β is the dimension of $\beta_t^{(u)}$ and $\beta_t^{(v)}$. We then specify Gaussian priors for $m_k(i)$ for $k = u, v$ and $i = 1, \dots, n_\beta$; Gaussian priors for the initial conditions, $\beta_0^{(u)}$ and $\beta_0^{(v)}$; and independent inverse gamma priors for all variances and the elements of λ . For details of the hyperparameter specifications, see Milliff et al. (2011).

9.4.3 The SVW BHM: Implementation and Inference

As discussed above, the eventual goal of this analysis is to use the SVW BHM to produce realistic realizations of wind fields to initialize the MFS operational ocean-forecast model. The BHM was implemented for various 24-day periods in 2005 and 2006, with the first 14 days in each corresponding to the “assimilation” period, where both scatterometer and ECMWF winds were available, and the last 10 days corresponding to a “forecast” period, where SVW observations and analyses are replaced by ECMWF surface-wind forecasts. That is, there are no “observations” in the forecast period, but the BHM treats the ECMWF forecasted SVWs as if they were data

during this period. The wind component processes were considered on a grid with $0.5^\circ \times 0.5^\circ$ spatial resolution and a 6-h time resolution. For BHM-implementation details, see Milliff et al. (2011) and Bonazzi (2008).

Figure 9.16 shows posterior-distribution summaries for the wind process for February 2, 2005, over the western portion of the Mediterranean basin. The red vector shows the posterior mean, and the variation about that mean is illustrated by the spread (i.e., variation) in 10 samples from the posterior distribution (black vectors), taken at widely separated intervals in the MCMC iterations. Each of the 11 vectors in a grid cell emanates from a common vertex; therefore, the *direction* of a vector is *away* from its vertex. Note that at any given grid cell, there is considerable spread in the black vectors, which could be important when these winds are used to force an ocean model. In particular, note that the so-called *Mistral winds* (strong northerly winds blowing off the south coast of France into the Gulf of Lyon) are quite evident, as well as strong north–northwest winds in the region of the Sicily Straits. In general, although there is reasonable spread associated with these high wind speeds, the wind (direction and speed) uncertainty is larger in more sheltered regions, where the wind speeds are lower on average.

The atmospheric winds force the ocean primarily through the wind-stress curl (see Section 7.4; Milliff and Morzel, 2001). Figure 9.17 shows the posterior mean and posterior standard deviation for wind-stress curl for the four model times on February 2, 2005. In this case, we simply show the standard deviations calculated from the 10 realizations (i.e., ensembles) of wind fields taken from the posterior, which are shown in Figure 9.16. Note that relatively small differences in the wind fields can lead to large differences in the wind-stress curl,

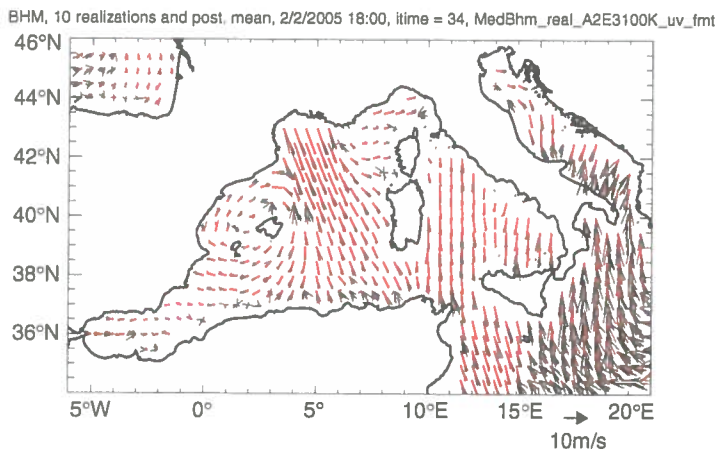


Figure 9.16 Posterior means (red vectors) and ten samples (black vectors) from the posterior distribution for SVW over the western Mediterranean basin for 18:00 UTC on February 2, 2005. Wind speed is proportional to the length of the vectors. The direction of a vector is away from the vertex at the center of its grid cell.

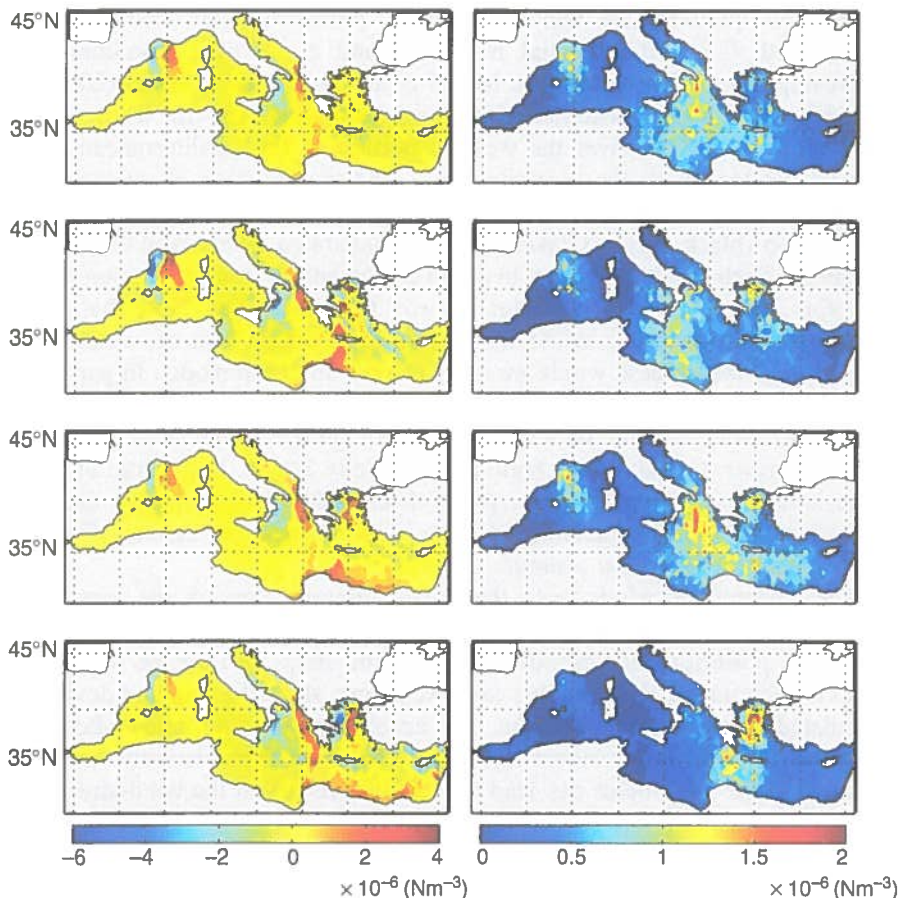


Figure 9.17 Wind stress curl maps for 00:00, 06:00, 12:00, and 18:00 UTC (from top to bottom) for February 2, 2005. The left panels show the posterior means obtained from the SVW BHM; the right panels show the corresponding standard deviations obtained from the 10 samples from the posterior distribution shown in Figure 9.16.

due to the nonlinear nature of the drag law used to derive this quantity. The dominant features in the wind-stress-curl posterior mean (e.g., corresponding to the Mistral event in the Gulf of Lyon) are at relatively large (i.e., “synoptic”) spatial scales, whereas the areas of greatest standard deviations tend to be at smaller (i.e., “mesoscale”) spatial scales.

The small-scale variations in wind forcing can have dramatic impacts on the ocean, in that they lead to uncertainties in the ocean state variables at comparable scales of spatial resolution. Uncertainties on these scales correspond to the scales of greatest uncertainty in the ocean-forecast system as well. This is exactly what the forecaster wants—knowledge in regions where the ocean model and analyses are least certain. The resulting uncertainties can

then be propagated through the deterministic ocean model to produce realistic forecast uncertainties. In particular, the ensembles from the SVW BHM have a substantial and realistic impact on the ocean initial conditions, because they are incorporated throughout a 14-day sequential data assimilation system in the MFS. For example, to see how the uncertainties in the initial wind forcing, obtained from the BHM, impact the multivariate ocean state variables, consider the impact on sea surface temperature (SST) and sea surface height (SSH). Figure 9.18 shows the posterior mean and posterior standard deviation for the MFS-derived SST initial condition for March 14, 2006 at 12:00 UTC, based on the corresponding SVW posterior distributions used in the previous 14-day data-assimilation cycle. Similarly, Figure 9.19 shows the posterior mean and posterior standard deviation for the corresponding SSH. In both cases, the posterior means exhibit relatively large-scale features of the general ocean circulation and its response to wind forcing. Importantly, the posterior standard deviations show that the uncertainties are at much smaller, but coherent spatial scales. Remarkably, as discussed in Milliff et al. (2011), these features correspond to persistent “hot spots” in the mesoscale circulation of the Mediterranean Sea (e.g., the Algerian Current, the Atlantic-Ionian Stream, and the Mersa Matruh gyre). Thus, the uncertainty in the initial wind

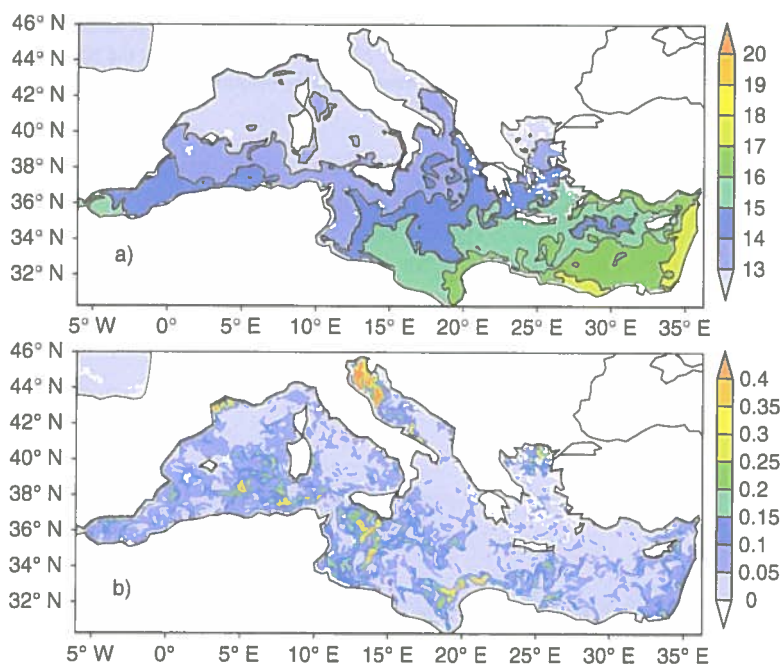


Figure 9.18 Sea surface temperature (SST, in units of °C) posterior means (top panel) and posterior standard deviations (bottom panel) at 12:00 UTC on March 14, 2006. Note that the posterior standard deviations were obtained from 10 realizations from the SST posterior distribution.

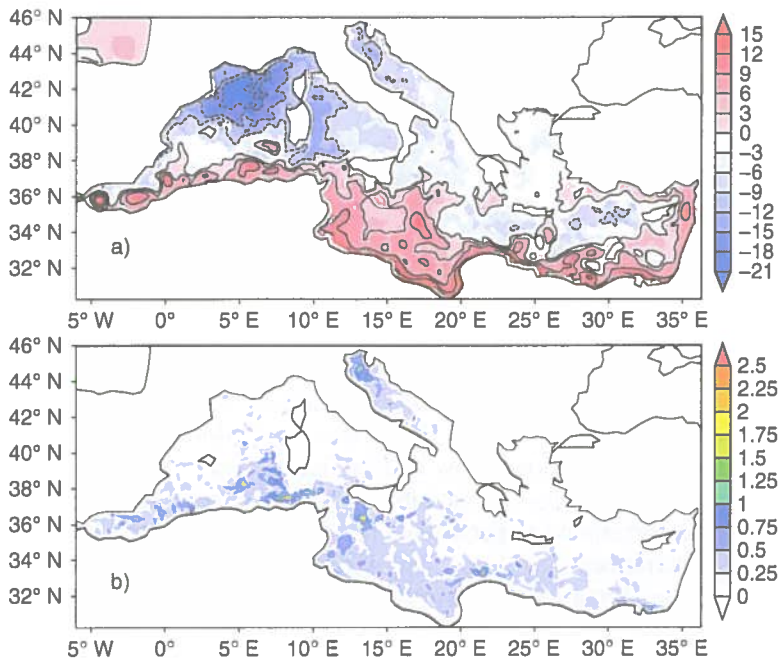


Figure 9.19 Sea surface height (SSH, in units of cm) posterior means (top panel) and posterior standard deviations (bottom panel) at 12:00 UTC on March 14, 2006. Note that the posterior standard deviations were obtained from 10 realizations from the SSH posterior distribution.

fields have provided a localization of forecast uncertainty in the oceanic state variables, which is essential for producing realistic forecast uncertainties in the operational ocean forecast system. Specific details regarding the impact of these winds from the SVW BHM, in the MFS operational ocean model, are given in Pinardi et al. (2011).